

PRIMARY 5 PREVOCATIONAL STUDIES SCHEME OF WORK

FIRST TERM

Week	Topic	Content
1	Soil Classification I	Types of soil - sandy, clay, loamy; Characteristics of each soil type; Identifying different soils; Soil texture and feel
2	Soil Classification II	Soil structure and composition; Soil color and what it indicates; Topsoil and subsoil; Soil profile; Soil testing
3	Soil Enrichment I	Meaning of soil enrichment; Why soil needs enrichment; Organic matter and humus; Composting basics; Animal manure
4	Soil Enrichment II	Fertilizers - organic and inorganic; Green manuring; Crop rotation; Cover crops; Improving soil fertility
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Agricultural Tools and Equipment I	Hand tools - hoes, cutlasses, rakes, spades, shovels; Functions of each tool; Differences and similarities; Selection of appropriate tools
8	Agricultural Tools and Equipment II	Simple machines in agriculture - wheelbarrow, irrigation equipment, sprayers; Watering cans and hose pipes; Measuring tools; Modern vs traditional tools
9	Safety Measures in Agriculture I	Personal protective equipment (PPE) - gloves, boots, hats, overalls; Importance of PPE; When to use protective gear
10	Safety Measures in Agriculture II	Safety precautions when using tools; Preventing accidents; First aid on the farm; Safe storage of tools and chemicals; Farm safety rules
11	REVISION	Review of all topics covered; Practice exercises; Question and answer sessions
12	EXAMINATION	End of term assessment

13	CLOSING	Report card distribution; Vacation assignment
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PRIMARY 5 PREVOCATIONAL STUDIES LESSON NOTES

WEEK 1: SOIL CLASSIFICATION I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List the three main types of soil.
2. Describe the characteristics of sandy, clay, and loamy soil.
3. Identify soil types by their texture (feel).
4. State the best soil for planting crops.

CONTENT

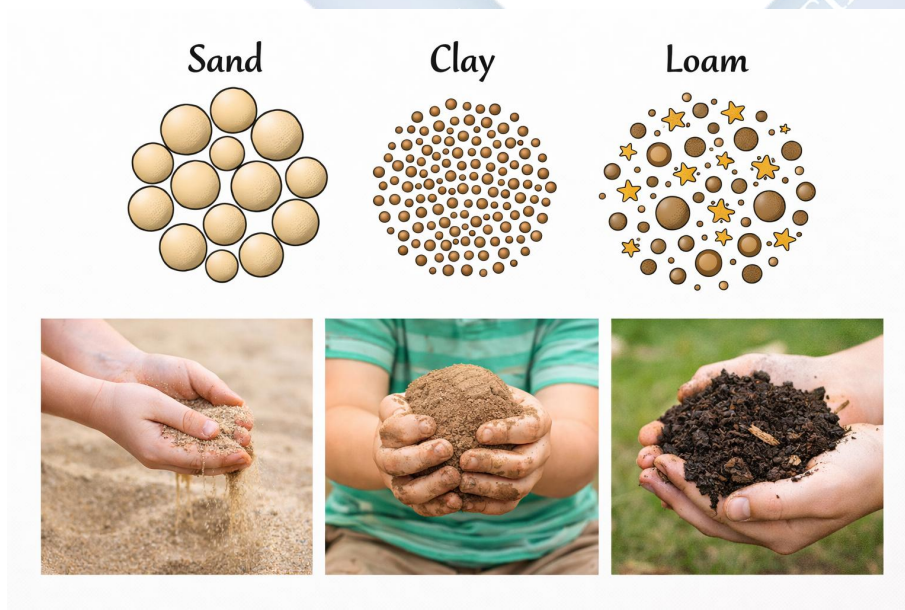
1. Types of Soil Soil is the **living skin of the earth**. It is the top layer where plants put down their roots to grow. In Nigeria, we have three main types of soil that you can find in different places:

- Sandy Soil
- Clay Soil
- Loamy Soil

2. Characteristics of Each Soil Type

- **Sandy Soil:**
 - **Look & Feel:** It is light in colour (often light brown or greyish) and feels **rough and gritty** when you rub it between your fingers, like the sand at the beach in Lagos or Calabar.
 - **Particles:** It is made of **large particles** with **big spaces** between them.
 - **Water & Air:** Water drains through it **very quickly** because of the big spaces. It cannot hold water well, so it dries out fast. Air moves through it easily.
 - **Where found:** Beaches, deserts, and some parts of our villages where nothing seems to grow well.
- **Clay Soil:**

- **Look & Feel:** It is often reddish, brown, or yellow. When **wet**, it feels **slippery and sticky**—you can mold it like play dough. When **dry**, it becomes **very hard and cracks**.
 - **Particles:** It is made of **very tiny, flat particles** that pack together tightly.
 - **Water & Air:** It holds onto water **too well**, becoming waterlogged. Air cannot pass through easily, so plant roots can suffocate.
 - **Where found:** In areas where water collects, like valleys or near rivers. Potters use it to make clay pots.
- **Loamy Soil:**
 - **Look & Feel:** It is **dark brown or black** and feels **soft, crumbly, and slightly damp**, like a rich chocolate cake. It is neither gritty nor sticky.
 - **Particles:** It is a **perfect mixture** of sand, clay, and lots of **humus** (decayed plants and animals).
 - **Water & Air:** It holds water **just right**—enough for plants but not too much. It also has plenty of air pockets. It is full of nutrients (plant food).
 - **Where found:** In good farms and gardens where plants grow green and healthy.



3. Soil Texture and Feel – The "Hand Test"

You can identify soil by doing a simple "Hand Test":

1. Take a small handful of slightly damp soil.
2. Rub it between your fingers.
3. Try to squeeze it into a ball.
 - **Sandy soil:** Feels gritty. The ball **crumbles immediately**.
 - **Clay soil:** Feels smooth and sticky. The ball holds its shape **perfectly** and can even be rolled into a sausage.
 - **Loamy soil:** Feels soft. The ball holds together **but breaks apart easily** when poked.

4. Best Soil for Planting

Loamy soil is the **best** for farming and gardening. It is often called "agricultural gold" because:

- It provides excellent **support** for roots.
- It holds the perfect amount of **water**.
- It is rich in **nutrients** from humus.
- It has good **aeration** (air) for roots to breathe. Most of our staple crops like maize, yam, and vegetables grow best in loamy soil.

EVALUATION

1. Which soil type would you find at the beach, and how would it feel in your hand?
2. Why would a plant struggle to grow in clay soil during the rainy season?
3. If you can roll soil into a long, thin sausage without it breaking, what type is it likely to be?
4. What gives loamy soil its dark brown colour and makes it so good for plants?
5. A farmer's soil dries out too fast and water runs straight through. Which soil type does he have, and what should he mix into it to improve it?
6. True or False: Sandy soil is the best for making pottery.

ASSIGNMENT

1. **Practical:** Collect two small soil samples from near your home (e.g., compound, garden path). Do the "Hand Test." In your notebook, describe how each one felt and guess its type.
2. **Drawing:** Draw three columns. Label them Sandy, Clay, Loamy. In each column, draw simple diagrams to show the size of the particles and the spaces between them.
3. **Oral:** Explain to a classmate why a gardener prefers loamy soil, using the words "water," "air," and "food for plants."



WEEK 2: SOIL CLASSIFICATION II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define soil profile.
2. Differentiate between topsoil and subsoil.
3. Explain what soil colour indicates.
4. Describe a simple way to test soil.

CONTENT

1. Soil Structure and Profile

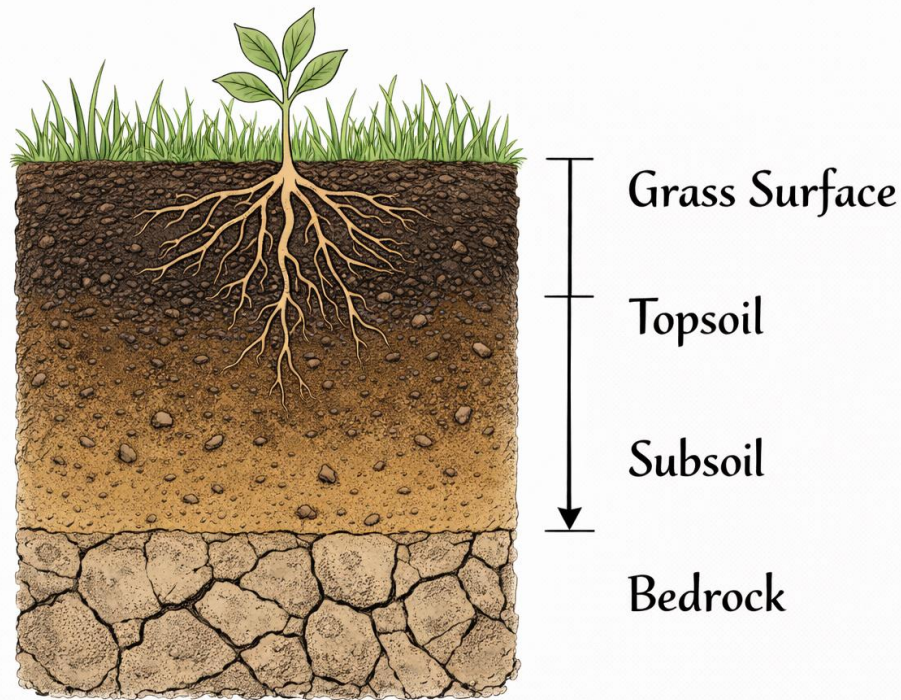
If you dig a deep hole, about the height of two rulers, you will see that soil is arranged in **layers**, like a cake. This vertical arrangement of layers is called the **Soil Profile**. The main layers from top to bottom are:

- **Topsoil (A-Horizon):** The top layer, rich and dark.
- **Subsoil (B-Horizon):** The middle layer, lighter and harder.
- **Parent Rock/Bedrock (C-Horizon):** The bottom layer, solid rock.

2. Topsoil vs. Subsoil

- **Topsoil:**
 - This is the **most important layer** for farming.
 - It is dark because it contains **humus**—rotting leaves, dead insects, and animal waste.
 - It is soft, loose, and full of life like earthworms and tiny insects.
 - **Most plant roots grow here** to find food and water.
- **Subsoil:**
 - This layer lies below the topsoil.
 - It is **lighter in colour** (brown, red, or yellow) because it has little or no humus.
 - It is **harder and more compacted**. It contains minerals washed down from the topsoil and small stones.

- Very deep roots of big trees reach down here.



3. Soil Colour and What It Indicates

The colour of soil is like a **message** about what is inside it.

- **Dark Brown/Black:** This is **good news!** It means the soil is rich in **organic matter (humus)** and is very fertile. Perfect for planting.
- **Red or Yellow:** This colour often comes from **iron minerals** (like rust). It usually means the soil is **well-drained** (water flows through it well). It can be good for crops but may need more organic matter.
- **Grey or Bluish:** This is a **warning sign**. It often means the soil is **waterlogged** (too much water, not enough air). Plants may drown here. It can also mean the soil lacks nutrients.

4. Simple Soil Testing – The "Jar Test"

Farmers test soil to understand what it's made of. We can do a simple **Jar Test**:

1. **Collect:** Get a clear glass jar or plastic bottle.
2. **Fill:** Put a handful of soil from your garden into the jar (about 1/4 full).
3. **Add Water:** Fill the jar almost to the top with water.
4. **Shake:** Close the lid tightly and shake hard for one minute.
5. **Observe:** Let the jar sit still for 24 hours. The soil will settle into layers.
 - **Bottom layer:** Heavy **sand** particles.
 - **Middle layer:** **Silt** (finer than sand).
 - **Top layer:** **Clay** particles.
 - **Floating on top:** **Organic matter** (humus).



EVALUATION

1. What do we call the different layers of soil we see when we dig a deep hole?
2. Which layer of the soil profile would an earthworm most likely live in, and why?
3. If a farmer's soil is dark black, what does that tell him about its fertility?
4. In the Jar Test, which soil particle is the heaviest and settles at the very bottom?
5. Why is the subsoil usually a lighter colour than the topsoil?
6. A patch of land has greyish soil and puddles that never dry. Is it good for planting okra?
Why or why not?

ASSIGNMENT

1. **Investigation:** With a parent's permission, look at a freshly dug hole (like for a fence post or construction). Try to identify the topsoil and subsoil by their colour. Describe what you see.
2. **Drawing:** Draw the soil profile you observed or the one from the lesson. Colour and label the layers (Topsoil, Subsoil, Bedrock).
3. **Written:** Write one reason why a farmer would want to do a Jar Test before planting a new crop.



WEEK 3: SOIL ENRICHMENT I

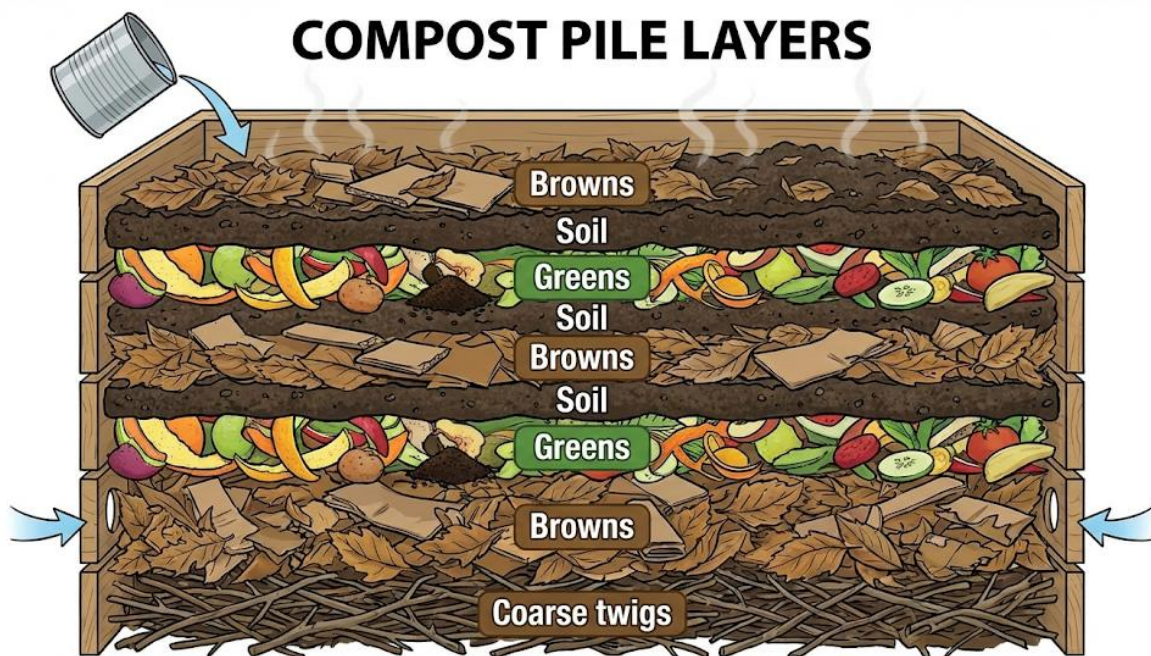
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define soil enrichment.
2. State two reasons why soil needs enrichment.
3. Explain the meaning of organic matter and humus.
4. Describe how to make simple compost.

CONTENT

1. Meaning of Soil Enrichment

Soil enrichment or **soil improvement** is the process of **adding good things back into the soil** to make it more fertile (richer) so that plants can grow better. Think of it like **feeding the soil** because hungry soil grows weak plants.



2. Why Soil Needs Enrichment (Becomes Hungry)

Soil loses its nutrients (plant food) for several reasons:

- **Crop Removal:** When we harvest maize, cassava, or vegetables, we are taking nutrients away from the soil. If we don't replace them, the soil becomes poor.
- **Leaching:** Heavy rain (like during the rainy season) can **wash nutrients deep down** into the ground, away from plant roots.
- **Erosion:** Wind and water can **carry away the rich topsoil**, leaving behind the poor subsoil.
- **Continuous Planting:** Planting the same crop on the same land year after year uses up the same nutrients.

3. Organic Matter and Humus: Soil's Superfood

- **Organic Matter:** This is anything that was once alive and is now decaying. Examples:
 - **Plant-based:** Dead leaves, grass cuttings, yam/plantain peels, maize stalks.
 - **Animal-based:** Animal manure (chicken droppings, cow dung), fish waste.
- **Humus:** This is the **final, dark, crumbly product** when organic matter has fully broken down by tiny organisms in the soil. **Humus is magic for soil! It:**
 - Holds water like a sponge.
 - Makes soil loose and airy.
 - Releases nutrients slowly to feed plants.

4. Composting Basics – Recycling Waste into Wealth

Compost is decayed organic matter used as a natural fertilizer. Making compost is like **cooking food for the soil**.

- **How to make a simple compost pile:**
 - a. **Choose a spot:** A corner of the garden.
 - b. **Start with a layer** of dry, brown materials (small twigs, dry leaves).
 - c. **Add a layer** of green materials (vegetable scraps, fresh grass).
 - d. **Add a thin layer** of soil or old compost.
 - e. **Repeat** the layers (brown, green, soil).
 - f. **Keep it moist** (like a damp sponge) and turn it with a stick every few weeks.

- **What happens:** Tiny bacteria, fungi, and earthworms will eat the waste! After 2-3 months, it turns into dark, sweet-smelling, crumbly **compost** ready for the garden.



EVALUATION

1. What is the main goal of soil enrichment?
2. How does harvesting crops make the soil "hungry"?
3. What is the difference between "organic matter" and "humus"?
4. Why is humus so valuable for a farmer's field?
5. List two "green" materials and two "brown" materials you could use for compost.
6. True or False: Plastic bags and bones are good materials for a compost pile.

ASSIGNMENT

1. **Survey:** List five items from your kitchen or compound that could be composted instead of thrown away (e.g., orange peels, eggshells).
2. **Practical:** With permission, start a very small compost experiment in a large pot or bucket. Add a few layers of soil, dry leaves, and fruit peels. Water it lightly and observe it weekly.
3. **Drawing:** Draw the cycle of compost: Kitchen/Garden Waste --> Compost Pile --> Rich Humus --> Healthy Plants.

WEEK 4: SOIL ENRICHMENT II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Differentiate between organic and inorganic fertilizers.
2. Define crop rotation.
3. Explain the use of cover crops.
4. Mention one benefit of green manuring.

CONTENT

1. Fertilizers: Organic vs. Inorganic

Fertilizers are materials added to soil to supply nutrients. They come in two main types:

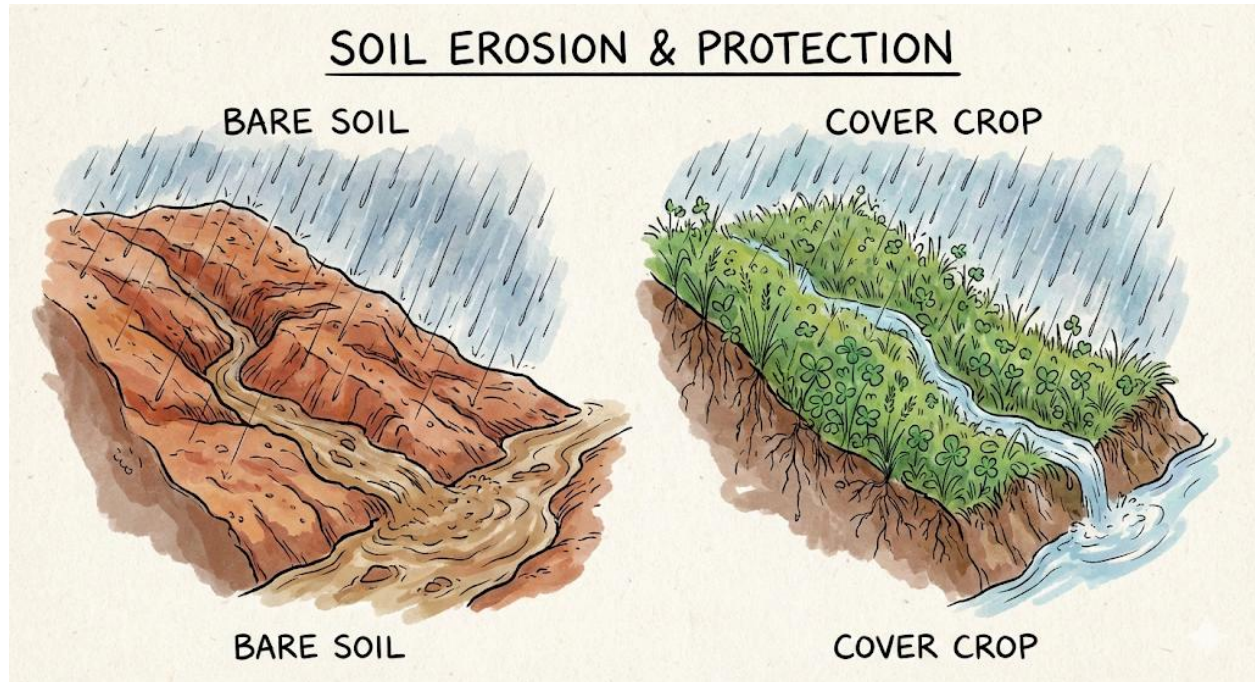
Feature	Organic Fertilizers	Inorganic Fertilizers
Source	From nature (plants & animals)	Made by humans in factories
Examples	Compost, Animal manure (chicken, goat), Bone meal	NPK 15:15:15, Urea, Superphosphate
How it works	Slow-release. Feeds soil life first, then plants. Improves soil structure.	Fast-release. Feeds plants directly. Works quickly.
Effect on Soil	Improves soil health in the long term.	Can damage soil if used too much (makes soil hard/acidic).
Cost	Often cheap or free to make.	Must be bought with money.

2. Crop Rotation – Giving the Soil a Rest

Crop rotation is a smart farming practice where a farmer grows **different types of crops on the same piece of land in a planned order over seasons or years.**

- **Why it works:** Different crops use and give different nutrients.
- **Example of a 2-year rotation:**
 - **Year 1:** Plant **Maize**. Maize uses up a lot of nitrogen from the soil.

- **Year 2:** Plant **Beans or Groundnuts** (legumes). These plants have special bacteria on their roots that **put nitrogen back into the soil**.
- **Result:** The soil stays balanced and fertile.



3. Cover Crops – A Living Blanket for the Soil

Cover crops are plants grown not mainly for harvest, but to **protect and improve the soil**.

- **Examples:** Melon, Cowpea, Sweet potato vines.
- **Benefits:**
 - Prevent Erosion:** Their leaves and roots hold the soil tight against heavy rain and wind.
 - Suppress Weeds:** They cover the ground, blocking sunlight from weeds.
 - Improve Soil:** When they are ploughed in, they add organic matter.

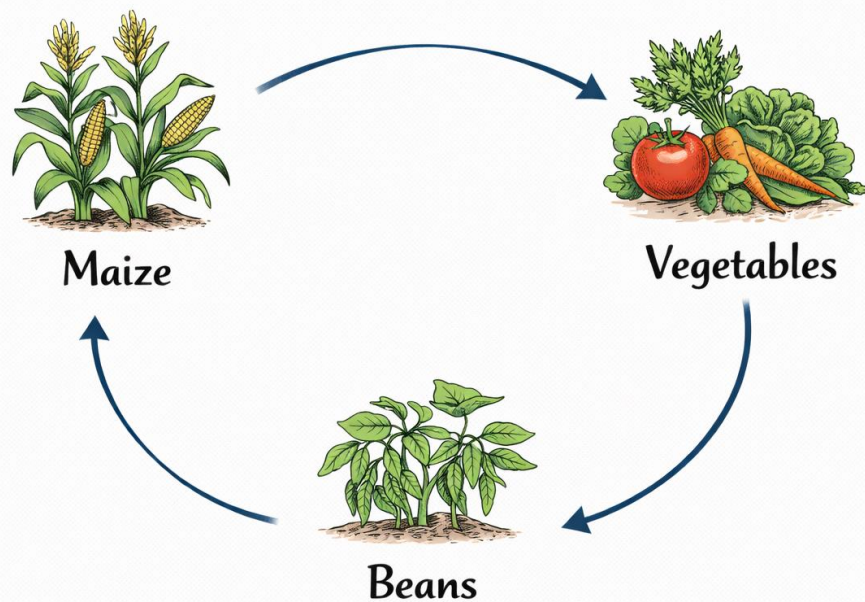
4. Green Manuring – A Fast Food for Soil

Green manuring is a type of cover cropping where fast-growing plants (like legumes) are grown and then **ploughed directly into the soil while they are still young and green**.

- **Process:** The green plants are buried in the soil where they rot quickly.

- **Benefit:** They release nutrients **very fast**, giving a quick boost to the next crop planted there.

3-Year Crop Rotation Cycle



EVALUATION

1. If a farmer wants to improve his soil structure for the next 5 years, should he use more organic or inorganic fertilizer? Why?
2. What does the "N" in NPK fertilizer stand for, and why is it important for plants?
3. Explain in your own words why planting beans after maize is a smart idea.
4. How does a cover crop like melon protect the soil during a heavy storm?
5. What is the main difference between a regular cover crop and a green manure?
6. True or False: Using only inorganic fertilizer every season is the best long-term plan for soil health.

ASSIGNMENT

1. **Written:** Design a simple 2-year crop rotation plan for a small family plot. Choose crops common in your area (e.g., Year1: Cassava, Year2: Okra/Beans).

2. **Investigation:** Look at a nearby farm or garden. Can you see any evidence of crop rotation, cover crops, or use of manure? Write down one observation.
3. **Drawing:** Draw a bag of NPK fertilizer. On the bag, write one advantage and one disadvantage of using it.



WEEK 5 & 6: MIDTERM EXAMINATION AND BREAK



WEEK 7: AGRICULTURAL TOOLS AND EQUIPMENT I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Identify common agricultural hand tools (hoe, cutlass, rake, spade, shovel).
2. State the function of each tool.
3. Differentiate between a spade and a shovel.
4. Mention why we need to select the right tool for work.

CONTENT

1. Common Agricultural Hand Tools

These are simple tools operated by human hands. They are essential for small-scale farming in Nigeria.

2. Functions and Uses of Tools

- **Cutlass (Machete):**
 - **Description:** A long, sharp metal blade with a wooden or plastic handle.
 - **Primary Use:** Clearing bushes, cutting tall grasses, and harvesting crops like sugarcane.
 - **How to use:** Swing carefully to cut vegetation at the base.
- **Hoe:**
 - **Description:** A metal blade (often rectangular or heart-shaped) fixed at an angle to a long handle.
 - **Primary Use:** Digging, turning soil, making ridges/heaps (mounds), and weeding.
 - **How to use:** Lift and drive the blade into the soil, then pull towards yourself.
- **Rake:**
 - **Description:** A long handle with a head made of metal or plastic teeth, like a comb.

- **Primary Use: Gathering** loose materials like dry leaves, grass, and stones. Also used for **levelling** soil after digging.
- **How to use:** Pull the teeth across the ground to collect debris.
- **Spade:**
 - **Description:** A **flat, nearly rectangular** metal blade with a slightly curved edge, attached to a handle.
 - **Primary Use: Digging straight-edged holes, turning soil, slicing through roots.** Used for precise digging and transplanting.
 - **How to use:** Push the blade vertically into the soil using your foot on the step (shoulder).
- **Shovel:**
 - **Description:** A **curved, scoop-shaped** metal blade with raised sides.
 - **Primary Use: Scooping and moving** loose materials like sand, gravel, manure, or harvested grains.
 - **How to use:** Slide the blade under the material and lift.



3. Key Difference: Spade vs. Shovel

Remember: A **spade is for digging** (it cuts and lifts soil). A **shovel is for scooping** (it lifts and carries loose stuff). You dig a planting hole with a **spade**, but you move the dug-out soil with a **shovel**.

4. Importance of Selecting the Right Tool

Using the correct tool makes work **easier, faster, and safer**.

- **Wrong Tool = Hard Work:** Trying to dig a hole with a rake is nearly impossible.
- **Wrong Tool = Danger:** Using a blunt or broken cutlass can cause accidents.
- **Right Tool = Good Result:** Using a hoe to make neat ridges gives seeds a perfect bed to grow in.

EVALUATION

1. Which tool would you use to clear a patch of land full of tough weeds and small shrubs?
2. A farmer needs to make long, straight mounds for planting yam. Which tool is best?
3. You need to move a pile of sand from one corner of the school garden to another. Do you choose a spade or a shovel? Why?
4. What is the main job of a rake on a farm?
5. Why is it important to use a sharp cutlass instead of a dull one?
6. If you see a tool with a flat, straight-edged blade, what is its likely name and main use?

ASSIGNMENT

1. **Drawing:** In your notebook, draw a **Hoe** and a **Cutlass**. Label three parts on each (e.g., Blade, Handle, Neck).
2. **Practical Survey:** At home, look for any of these tools. Ask an adult its name and one thing they use it for. Report your findings.
3. **Matching:** Create a simple table matching the tool to its job: Tool: (Cutlass, Hoe, Rake) | Job: (Levelling soil, Clearing bush, Making ridges).

WEEK 8: AGRICULTURAL TOOLS AND EQUIPMENT II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Identify simple farm machines (wheelbarrow, watering can).
2. Explain the use of a sprayer.
3. Mention tools used for measuring on the farm.
4. Compare traditional and modern tools.

CONTENT

1. Simple Farm Machines (Implements) These tools help to make farm work less stressful and more efficient.

- **Wheelbarrow:**
 - **Description:** A small cart with **one wheel** at the front, two handles, and a metal or plastic basin.
 - **Use: Transporting** heavy or bulky loads like harvested crops, manure, compost, or tools from one place to another. It uses the principle of a lever to make lifting easier.
- **Watering Can:**
 - **Description:** A portable container with a handle, a spout, and a **rose** (a perforated metal head with many small holes).
 - **Use: Watering** seedlings, young plants, or vegetables in a garden. The rose breaks the water into a gentle "rain" so it doesn't wash away the soil or damage delicate plants.
- **Hose Pipe:**
 - **Description:** A long, flexible rubber or plastic tube.
 - **Use: Carrying water** from a tap or tank to water plants over a larger area without carrying cans.



2. Sprayers and Irrigation

- **Knapsack Sprayer:**

- **Description:** A tank carried on the back like a school bag, with a pump handle, hose, and nozzle.
- **Use:** **Spraying** liquid chemicals evenly over crops. It can be used for:
 - **Pesticides:** To kill insect pests.
 - **Herbicides:** To kill weeds.
 - **Fungicides:** To control plant diseases.
- **Safety Note:** The user must wear gloves, a mask, and cover their skin.

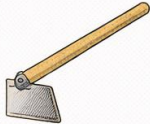
- **Simple Irrigation:** This is artificial watering. A **sprinkler** is a device that sprays water in a circle like rain.

3. Measuring Tools Farming needs planning and measurement.

- **Measuring Tape:** For measuring land size, plot boundaries, or distances between plants (spacing).
- **Ranging Poles:** Long, straight poles painted in alternating red and white bands. They are used in pairs to mark **straight lines** for planting or digging trenches.

4. Traditional vs. Modern Tools

- **Traditional Tools:** Hoe, cutlass, watering can. They are **cheaper, simpler**, and use human power. Best for **small farms** and gardens.
- **Modern Machines:** Tractors, ploughs, combine harvesters. They are **very expensive** but use engine power (fuel). They work **very fast** and are essential for **large-scale farming**.

Traditional	Modern
	
 Goggles	 Stover

EVALUATION

1. Why does a wheelbarrow have only one wheel at the front?
2. What is the purpose of the "rose" on a watering can?
3. What protective clothing should someone wear when using a knapsack sprayer with chemicals?
4. If a farmer wants to plant maize in straight rows, which measuring tool would help him mark the lines?
5. What is the main advantage of a tractor over a hoe for land preparation?

6. True or False: A hose pipe is more efficient for watering a large farm than a single watering can.

ASSIGNMENT

1. **Drawing:** Draw a well-labelled diagram of a wheelbarrow. Show the wheel, handles, and basin.
2. **List:** Name three things a farmer in your community might carry in a wheelbarrow.
3. **Reasoning:** Write one sentence explaining why a small-scale vegetable farmer might not need to buy a tractor.



WEEK 9: SAFETY MEASURES IN AGRICULTURE I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define Personal Protective Equipment (PPE).
2. List four examples of PPE used in agriculture.
3. Explain the importance of wearing boots and gloves.
4. State when to use a nose mask.

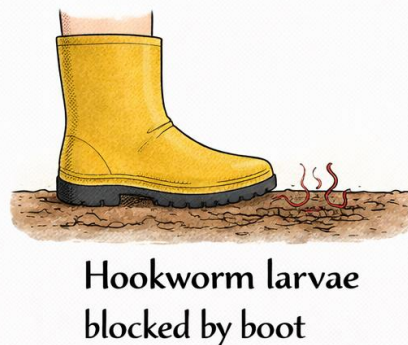
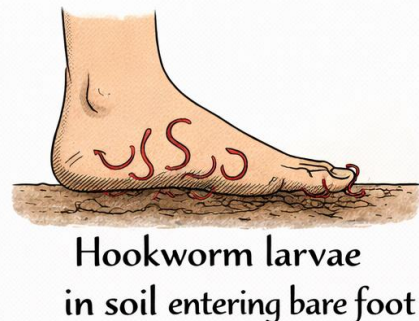
CONTENT

1. Personal Protective Equipment (PPE) - Your Farming Armour PPE stands for **Personal Protective Equipment**. Think of it as **special armour or a uniform** that farmers wear to protect their bodies from injuries, germs, and dangerous chemicals while working on the farm. Just like a footballer wears shin guards, a farmer needs protective gear.

2. Examples of PPE and Their Jobs

- **Hand Gloves:**
 - **Types:** Leather gloves (for handling rough tools, firewood), Rubber gloves (for mixing chemicals, cleaning).
 - **Protection:** They protect hands from **blisters, cuts, thorns, and chemical burns**. They also keep hands clean.
- **Rubber Boots (Rain Boots/Waterproof Boots):**
 - **Protection:** These are strong boots that cover your feet and ankles. They protect against:
 - **Sharp objects:** Nails, broken bottles, thorns hidden in grass.
 - **Dangerous animals:** Snake bites, scorpion stings.
 - **Infections:** Hookworm and other parasites that can enter through bare feet in damp soil.
- **Overalls or Coverall:**
 - **Description:** A long, thick one-piece suit that covers your regular clothes.

- **Protection:** Keeps your good clothes clean from mud, stains, and plant sap. Protects skin from scratches from branches and irritating plants.
- **Sun Hat/Cap with a Brim:**
 - **Protection:** Shields your head, face, and neck from the **hot sun**, preventing sunstroke and heat exhaustion. A wide brim is best.
- **Nose Mask/Respirator:**
 - **Protection:** Covers your nose and mouth.
 - **Use:** Worn when **spraying chemicals** (pesticides, herbicides) or working in **very dusty** conditions to prevent breathing in poisonous or harmful particles.
- **Goggles/Safety Glasses:**
 - **Protection:** Clear, protective glasses that shield your eyes from:
 - Chemical splashes from sprayers.
 - Dust, soil, or tiny wood chips flying while cutting or digging.



3. Importance of Key Protective Gear

- **Boots are Non-Negotiable:** Walking barefoot on a farm is one of the most dangerous things you can do. It's an open invitation to injury and sickness. **Always wear boots.**
- **Gloves Save Your Hands:** Your hands are your most important tools. Gloves prevent painful cuts that can get infected, and they stop harsh chemicals from causing rashes or sores.

4. When to Use Specific PPE It's not enough to have PPE; you must know **when** to use it.

- Use a **Nose Mask + Goggles + Gloves** when you are **mixing or spraying any chemical** on the farm.
- Use **Boots and Gloves** when you are **clearing land, weeding, or handling rough materials.**
- Use a **Sun Hat** any time you are working under the sun for a long time.

EVALUATION

1. What does PPE stand for, and why is it like a farmer's armour?
2. A farmer is about to spray weed killer. List three pieces of PPE he must wear and explain why for each.
3. Why is it more dangerous to walk barefoot on a farm than on your sitting room floor?
4. What could happen if you get a small cut on the farm and don't wear gloves?
5. Which PPE protects your eyes from flying particles, and when would you need it?
6. True or False: A sun hat is only for fashion on the farm.

ASSIGNMENT

1. **Drawing:** Draw a safety poster for your school farm. The poster should show a pupil dressed correctly in at least FOUR items of PPE. Label each item.
2. **Survey:** Ask a farmer, gardener, or construction worker in your community what protective clothing they wear. Write down two items they mentioned.
3. **Role-play:** With a partner, act out a short scene. One person is a farmer about to do a risky job (like clearing a bushy area). The other person is the "Safety Officer" who must tell them what PPE to wear and why.

WEEK 10: SAFETY MEASURES IN AGRICULTURE II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List three safety precautions when using tools.
2. Explain how to store farm tools safely.
3. Define First Aid.
4. State two general rules for farm safety.

CONTENT

1. Safety Precautions When Using Tools Using tools correctly prevents accidents.

- **Inspect Before Use:** Always check a tool before you pick it up.
 - Is the **handle loose or cracked**? A loose axe head can fly off!
 - Is the **blade sharp and clean**? A dull tool requires more force and is more likely to slip.
- **Focus and Concentration:**
 - **Look at what you are cutting or digging.** Never look away or talk distractedly while swinging a cutlass or hoe.
 - **Keep a Safe Distance:** Always be an arm's length plus the tool's length away from other people. No playing or running near someone using a tool.
- **Carry Tools Safely:** Carry sharp tools like cutlasses and hoes with the **blade pointing down** and away from your body. Never run with them.

2. Safe Storage of Tools and Chemicals

- **Clean and Dry:** After use, wash off mud and soil from tools. **Dry them completely** before storing to prevent rust.
- **Proper Storage:**
 - Hang tools on a rack or wall.
 - If they must be on the floor, lay them flat with the blade facing a wall. **Never leave a hoe or rake with the tines/blade pointing up**—someone could step on it.
 - Keep your tool storage area tidy.

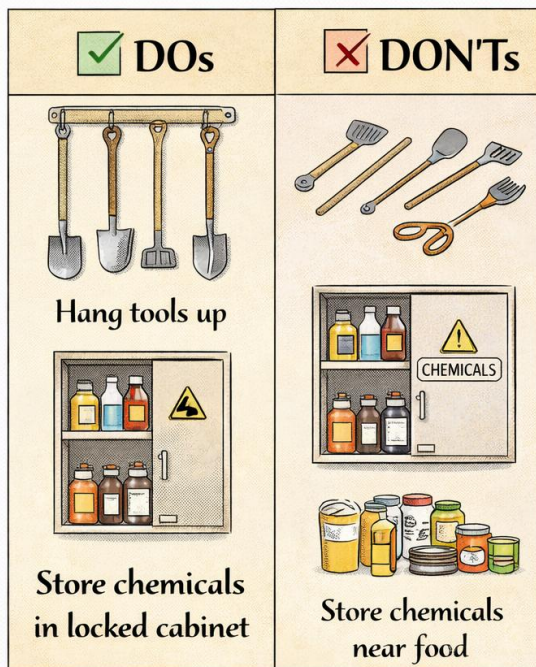
- **Chemical Safety:** Fertilizers and pesticides are **poisonous**.
 - **Store** in their **original containers** with labels clear.
 - **Keep** in a **locked cupboard** or high shelf, far away from children, food, and animal feed.
 - **Never put** chemicals in empty drink bottles (someone might drink it!).

3. First Aid on the Farm

- **Definition: First Aid** is the **immediate and simple help** given to someone who is injured or suddenly ill **before a doctor or nurse arrives**. Its main goals are to **preserve life, prevent the condition from worsening, and promote recovery**.
- **The First Aid Box:** Every farm should have one. It contains:
 - **Antiseptics:** Iodine, Methylated spirit (to clean wounds and kill germs).
 - **Dressings:** Cotton wool, gauze, bandages, plasters (to cover wounds).
 - **Tools:** Scissors, safety pins, tweezers (for removing splinters).
 - **Others:** Soap, clean water, gloves.

4. General Farm Safety Rules for Children

1. **Never Work Alone:** Always have an adult nearby.
2. **Do Not Taste or Eat:** Do not eat any unknown fruits, seeds, or berries you find on the farm.
3. **Beware of Water:** Do not drink from streams or ponds on the farm; the water may be contaminated.
4. **Report Immediately:** Tell an adult about **any** injury, broken tool, or spilled chemical right away.
5. **Watch Your Step:** Be careful on uneven ground to avoid tripping or twisting your ankle.



EVALUATION

1. Why is a tool with a loose handle more dangerous than one that is firmly fixed?
2. What is the safest way to carry a cutlass when walking from the shed to the farm?
3. Why is it extremely dangerous to store kerosene or pesticide in an empty soft drink bottle?
4. What is the main purpose of First Aid?
5. If you get a small cut on the farm, what are the first two things you should do (using the First Aid box)?
6. State two rules you must follow if you are allowed to visit a farm.

ASSIGNMENT

1. **Practical:** With an adult's help, check one tool at home (like a cutlass, hoe, or kitchen knife). Is the handle tight? Is the blade sharp? Write down your observation.
2. **Design:** Draw and label a "SAFE TOOL STORAGE" area for a small garden. Show where tools should be hung and where chemicals should be locked away.

3. **Creative:** Create a "Farm Safety Pledge" in your notebook with three promises you will keep for safety (e.g., "I promise to always wear shoes on the farm.").



WEEK 11: REVISION

CONTENT

1. Comprehensive Review of First Term Topics Let's revisit the key ideas we've learned this term:

- **Soil (Weeks 1-4):**
 - **Three Types:** Sandy (gritty, drains fast), Clay (sticky, holds water), Loamy (crumbly, best for planting).
 - **Soil Profile:** Layers of soil (Topsoil-dark/rich, Subsoil-lighter/harder).
 - **Soil Improvement:** Adding compost, manure, practicing crop rotation.
- **Farm Tools (Weeks 7-8):**
 - **Hand Tools:** Cutlass (clear), Hoe (dig/weed), Rake (gather/level), Spade (dig), Shovel (scoop).
 - **Machines/Equipment:** Wheelbarrow (carry), Watering can (water), Sprayer (spray chemicals).
- **Farm Safety (Weeks 9-10):**
 - **PPE:** Boots, Gloves, Hat, Mask, Goggles.
 - **Tool Safety:** Inspect, focus, carry safely, store properly.
 - **First Aid:** Immediate care; a First Aid box contains antiseptics, bandages, etc.

2. Interactive Revision Activities

- **"What Am I?" Game:** Teacher or pupil describes a tool or soil type without naming it. Others guess.
 - *Example:* "I am dark, crumbly, and hold water just right. Plants love me. What am I?" (Answer: Loamy soil).
- **Picture Quiz:** Show pictures of tools or PPE. Pupils name them and state one use or importance.
- **Safety Scenario Discussion:**
 - "Musa wants to help his father spray the crops. What should he wear?"

- "A hoe is left lying in the grass with the metal part facing up. What is the danger?"

EVALUATION *This session will be an informal, teacher-led assessment to gauge understanding before the termly examination.*

1. Describe how you would test a soil sample to see if it is sandy or clay.
2. Explain why a farmer might plant beans in a field that grew maize last season.
3. You need to make a new garden bed. Which tool would you use to break the soil and make ridges?
4. If you saw a classmate about to use a cutlass with a cracked handle, what would you tell them?
5. What should you **never** do if a small fire starts in a pot of oil on a stove?
6. Why is a First Aid box important in a place like the school farm or garden?

ASSIGNMENT

1. **Study Skills:** Create five revision flashcards. On one side, write a key term (e.g., "Loam," "Hoe," "PPE"). On the other side, write the definition or main use.
 2. **Practical Review:** Choose one topic from this term (Soil, Tools, or Safety). Teach one main point from it to a younger sibling or a friend. Write down what you taught them.
 3. **Organizational:** Ensure all your notes for Weeks 1-10 are complete, neat, and in order. Write one question you still have about any topic to ask your teacher.
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PRIMARY 5 PREVOCATIONAL STUDIES SCHEME OF WORK

SECOND TERM

Week	Topic	Content
1	Maintenance of Agricultural Tools and Equipment I	Meaning of maintenance; Importance of tool maintenance; Cleaning tools after use; Removing dirt and rust
2	Maintenance of Agricultural Tools and Equipment II	Sharpening tools; Oiling metal parts; Replacing broken handles; Proper storage to prevent damage; Regular inspection
3	Vocational Skills in Agriculture I	Meaning of vocational skills; Crop production skills - land preparation, planting, weeding, harvesting; Practical skills development
4	Vocational Skills in Agriculture II	Animal rearing skills; Processing agricultural products; Record keeping; Marketing farm produce; Farm management basics
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Food Classification and Functions I	Classification of food - carbohydrates, proteins, fats and oils; Sources of each food class; Functions in the body
8	Food Classification and Functions II	Vitamins and minerals; Water and roughage; Balanced diet; Food groups; Planning balanced meals
9	Basic Principles in Home Economics I	Principles of cooking - boiling, frying, roasting, steaming; Safety in food preparation; Hygiene in the kitchen
10	Basic Principles in Home Economics II	Principles of sewing - stitches, seams, hems; Basic hand stitching; Principles of cleaning; Home management principles
11	REVISION	Review of all topics covered; Practice exercises; Question and answer sessions
12	EXAMINATION	End of term assessment

13	CLOSING	Report card distribution; Vacation assignment
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PRIMARY 5 PRE-VOCATIONAL STUDIES LESSON NOTES

SECOND TERM

WEEK 1: MAINTENANCE OF AGRICULTURAL TOOLS AND EQUIPMENT I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define maintenance.
2. State three reasons why it is important to maintain tools.
3. Describe how to clean tools after use.
4. Explain how to remove rust from metal tools.

CONTENT

1. Meaning of Maintenance

Maintenance means **taking care of something to keep it in good working condition**. For farm tools, it is like giving them a **bath, a check-up, and a treatment** so they don't get sick (break down) and can work for a long, long time. Just as we brush our teeth to keep them healthy, we must care for our tools.

2. Why Tool Maintenance is Non-Negotiable

Taking care of tools is not an extra job; it is a **necessary job**. Here's why:

- **Longer Life (Durability):** A well-maintained hoe or cutlass can last for **years**, even decades, and be passed from parents to children. Neglected tools break quickly.
- **Efficiency and Ease:** A **sharp, clean tool** makes work **easier and faster**. A blunt, dirty hoe makes digging feel like punishment—it takes more energy and time.
- **Safety First:** A tool in good condition is a **safe tool**. A loose handle can cause the blade to fly off. A rusty, weak metal can snap unexpectedly. Maintenance prevents accidents.
- **Saves Money:** Fixing a small problem (like tightening a handle) is cheap. Letting it break completely means spending a lot of money to buy a new tool. "A stitch in time saves nine."

3. Cleaning Tools After Use – The Three-Step Rule

After a day's work, tools are tired, dirty, and sweaty. They need to be cleaned before resting (storage).

1. **Scrape Off Mud:** Use a stick, stone, or the edge of another tool to **knock off** big lumps of wet soil and mud. Do this while the mud is still damp, before it hardens like cement.
2. **Wash with Water:** Use water (from a bucket or hose) to **rinse off** the remaining dirt. For very stubborn sap or dirt, use a stiff brush (like a washing brush) with a little soap.
3. **Dry Thoroughly:** This is the **most important step!** Use an old rag, cloth, or sack to **wipe the tool completely dry**. Pay special attention to the metal blade and the area where the metal joins the wood. **Never** put a wet tool away. Water is the enemy of metal and wood.



4. Dealing with the Enemy:

Rust Rust is the reddish-brown, flaky coating that attacks iron and steel when they are exposed to air and moisture for too long. It is like a **sickness that eats metal**, making it weak and brittle.

- **How to Remove Rust:**

- a. **Scrub:** Use a piece of **rough sandpaper**, a scrubbing pad, or even dry sand/soil to scrub the rusty area vigorously until the brown flakes are gone and you see shiny metal.
- b. **Wipe:** After scrubbing, wipe the metal clean with an oily rag (a cloth with a few drops of palm oil, engine oil, or Vaseline). This oil layer will protect it from new rust.

EVALUATION

1. What does it mean to "maintain" a farm tool?
2. Explain how maintaining a tool can actually save a farmer money.
3. Why is it crucial to dry a tool completely before storing it?
4. What causes rust to form on a cutlass left outside in the rain?
5. Describe one safe method for removing rust from a garden trowel.
6. True or False: A blunt cutlass is safer than a sharp one because it's less likely to cut you.

ASSIGNMENT

1. **Practical:** Find one metal tool at home (e.g., kitchen knife, scissors, garden trowel). Clean it following the three-step rule (Scrape/Wash/Dry). Write a sentence about how it looked before and after.
2. **Drawing:** Draw a "Wanted" poster for "Rusty the Tool Destroyer." Describe what rust is and how to defeat it (by cleaning and oiling).
3. **Oral:** Explain to a family member two reasons why we shouldn't just throw tools in a corner after using them.

WEEK 2: MAINTENANCE OF AGRICULTURAL TOOLS AND EQUIPMENT II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Explain why we sharpen tools.
2. Describe the process of oiling metal parts.
3. State what to do with broken handles.
4. Mention two rules for proper storage of tools.

CONTENT

1. Sharpening Tools – Restoring the "Bite"

A **sharp tool** is an **effective and safe tool**. A blunt tool is dangerous because you have to use excessive force, which can lead to loss of control.

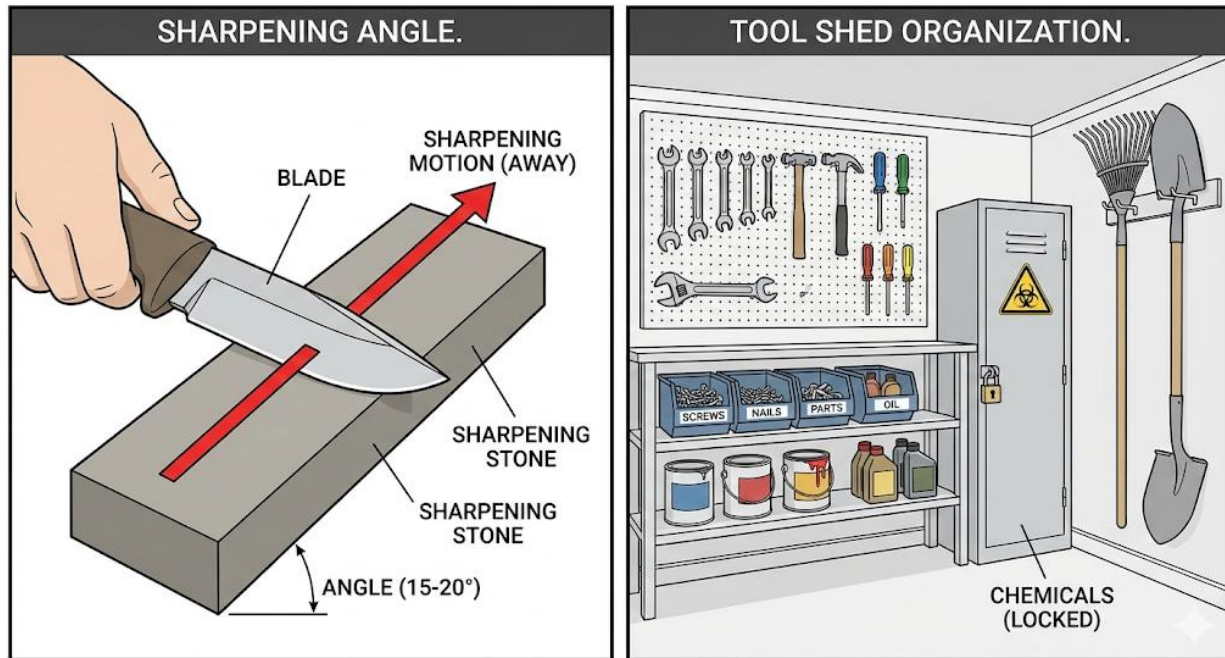
- **Why Sharpen?** Sharpening restores the **cutting edge**, so the tool slices through material (grass, soil, wood) with minimal effort.
- **Tools for Sharpening:**
 - **Sharpening Stone/File:** A rough, rectangular stone or a metal file. Used for hoes, cutlasses, axes.
 - **Whetstone:** A finer stone for putting a very sharp edge on knives.
- **Safe Sharpening Technique (for older children with supervision):**
 - a. Secure the tool firmly.
 - b. Hold the stone or file at the same angle as the existing edge.
 - c. Push the stone **away from your body** along the edge in smooth strokes.
 - d. Repeat until the edge is uniform and sharp. Test carefully.

2. Oiling Metal Parts – Creating a Protective Shield

Oiling is like putting **lotion on dry skin** or a **raincoat on metal**. It prevents air and moisture from touching the metal directly.

- **What to Use:** A light coating of **palm oil, engine oil, or grease**. Even Vaseline works.

- **How to Apply:** Put a few drops on a rag and **wipe it over all clean, dry metal parts.** Pay attention to joints and hinges. The thin layer keeps rust away during storage.



3. Fixing Handles – Don't Manage, Mend!

Wooden handles can crack, loosen, or break.

- **Loose Handle:** If the handle is wobbly, you can often fix it by **driving a small wedge** (a thin piece of wood or metal) into the top of the tool's eye (the hole where the handle goes). This tightens the fit.
- **Broken Handle: Do not tape it or try to use it!** A broken handle can snap completely and cause injury. The safe solution is to **replace it**. Remove the old handle and fit a new, strong one of the correct size.

4. Proper Storage and Regular Inspection

- **Storage Rules:**
 - a. **Dry Place:** Store in a shed, under a shelter, or inside. Never leave tools outside.
 - b. **Off the Ground:** Hang tools on racks or walls. If on a shelf, lay them flat.
 - c. **Organised:** Keep tools in order so you can find them easily and they don't bang into each other.

- **Inspection:** Make it a habit to **look over your tools** every few weeks. Check for loose parts, cracks, or early signs of rust. Fixing a small problem early stops it from becoming a big one.

EVALUATION

1. Why is a blunt cutlass considered more dangerous than a sharp one?
2. What is the main purpose of applying oil to a spade before storing it for the dry season?
3. Your friend is using a hoe with a slightly cracked handle. What advice should you give them and why?
4. Describe one safe and proper way to store a rake.
5. What should you look for during a routine tool inspection?
6. True or False: It is better to leave a slightly wet tool in the sun to dry than to wipe it with a cloth.

ASSIGNMENT

1. **Investigation:** With an adult's supervision, safely examine a cutlass or knife at home. Is the edge sharp or blunt? Report your finding. (Do not attempt to sharpen it without an adult).
2. **Design:** Draw a plan for a simple tool storage rack that can be made with wood. Show at least three tools hanging on it.
3. **Written:** Write a short "Maintenance Checklist" for a farmer with three items: 1. Clean after use. 2. _____. 3. _____. (Fill in the blanks with ideas from the lesson).

WEEK 3: VOCATIONAL SKILLS IN AGRICULTURE I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define vocational skills.
2. List three skills needed for crop production.
3. Describe the skill of weeding.
4. Mention why harvesting skills are important.

CONTENT

1. Meaning of Vocational Skills

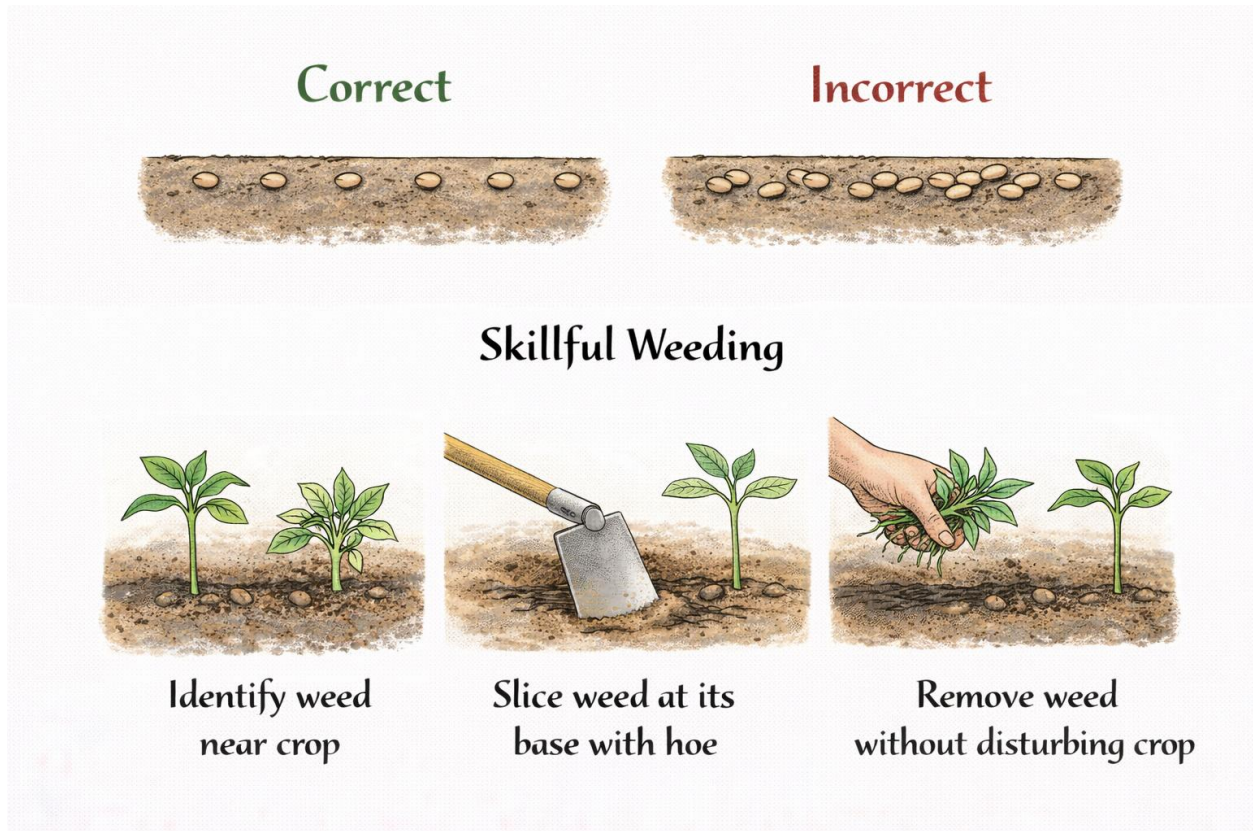
Vocational skills are **practical, hands-on abilities** that allow a person to do a specific job or task very well. In agriculture, these are not just about knowledge, but about **knowing how to do things correctly and efficiently** to produce food, raise animals, and make a living. It's about being a **skilled farmer**, not just a labourer.

2. Essential Crop Production Skills

Growing crops successfully requires a sequence of skilled actions:

- **Skill 1: Land Preparation**
 - This is **setting the stage** for planting. It involves:
 - **Clearing:** Skillfully using a cutlass to remove bushes without damaging valuable young trees.
 - **Stumping:** Removing tree stumps.
 - **Tilling/Ploughing:** Breaking and turning the soil to make it soft and aerated. Knowing how deep to plough for different crops.
 - **Making Ridges/Heaps:** Creating raised beds for crops like yam, cassava, and potatoes to improve drainage and yield. The ridges must be straight, evenly spaced, and firm.
- **Skill 2: Planting**
 - This is not just dropping seeds. It involves:
 - **Seed Selection:** Choosing healthy, disease-free seeds.

- **Correct Spacing:** Knowing the right distance between seeds (e.g., maize needs more space than beans). Proper spacing prevents competition for light and nutrients.
- **Correct Depth:** Planting seeds at the right depth—not too shallow (they dry out) and not too deep (they can't sprout).



3. The Skill of Weeding

- **Weeding** is removing **unwanted plants** (weeds) that compete with crops for water, sunlight, and nutrients.
- **Skillful Weeding Involves:**
 - **Identification:** Knowing which plants are weeds and which are young crops.
 - **Timing:** Weeding **early and regularly** is easier and more effective than waiting until weeds are big.

- **Method:** Knowing whether to **uproot** by hand (for small weeds) or use a **hoe** (for larger areas). The skill is to remove the weed **without damaging** the delicate roots of the crop nearby.
- **Tool Handling:** Using the hoe with a slicing motion close to the ground to cut weeds, not just chop at them.

4. Harvesting Skills – Reaping the Rewards Carefully

Harvesting is the **final and most rewarding** skill.

- **Importance of Skill:** Careless harvesting can **destroy** the hard work of months.
- **Key Skills:**
 - **Knowing Ripeness:** Recognizing the precise signs that a crop is ready. (e.g., Maize: dry, brown silk; Tomato: full, firm, and red).
 - **Using Correct Tools:** Using a sharp knife for fruits, a digger for tubers, to avoid bruising.
 - **Gentle Handling:** Treating the produce gently to prevent cuts, bruises, or crushing, which leads to quick spoilage.
 - **Timing:** Harvesting at the right time of day (often early morning) for best quality.

EVALUATION

1. How are "vocational skills" different from just "knowledge" about farming?
2. Why is it important to make ridges straight and evenly spaced?
3. What are two negative things that happen if a farmer does not weed his farm?
4. Describe one way a farmer can damage his tomato crop during harvest.
5. Why is it better to weed when the weeds are still small?
6. True or False: Any seed is good for planting as long as you put it in the soil.

ASSIGNMENT

1. **Practical/Observation:** Visit a home garden or school farm. Observe and write down one example of a vocational skill you see being used (e.g., someone weeding, planting, harvesting).

2. **Research:** Ask a gardener or farmer: "What is the sign you look for to know when your okra/pepper is ready to harvest?" Write down their answer.
3. **Drawing:** Draw a simple comic strip with three panels showing: 1. A seed planted correctly. 2. A plant growing strong. 3. The plant being harvested carefully.



WEEK 4: VOCATIONAL SKILLS IN AGRICULTURE II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List two skills needed for animal rearing.
2. Explain what it means to process agricultural products.
3. State the importance of record keeping.
4. Describe farm marketing.

CONTENT

1. Animal Rearing Skills

Raising animals (poultry, goats, fish) is called **animal husbandry**. It requires responsibility and specific skills.

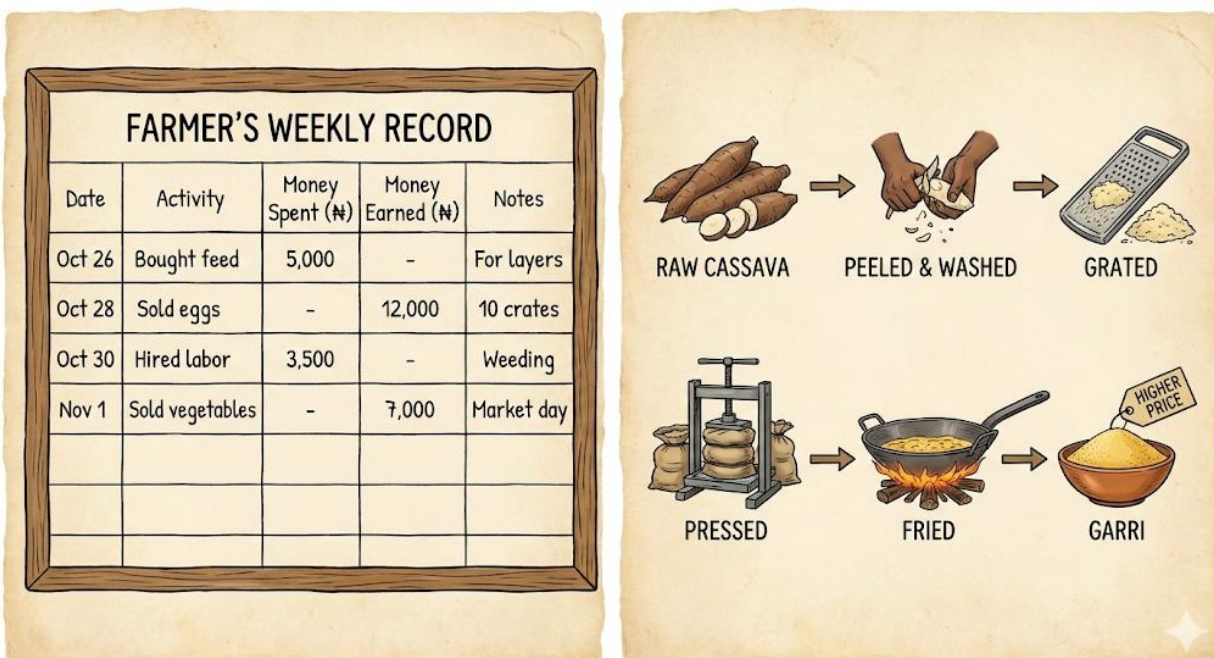
- **Skill 1: Feeding and Nutrition:**
 - **Knowledge:** Knowing the **right feed** for each animal at each stage of life (chick starter, grower mash, layers mash).
 - **Skill:** Providing feed **regularly and in correct quantities**. Knowing how to mix feed supplements if necessary.
- **Skill 2: Health Care and Management:**
 - **Observation:** Skillfully watching animals daily to spot signs of sickness early (e.g., dull eyes, drooping wings, not eating, isolation from the group).
 - **Hygiene:** Maintaining clean pens, coops, or ponds to prevent disease.
 - **Basic Treatment:** Knowing how to administer basic treatments or vaccines (with guidance) and when to call a veterinary doctor.

2. Processing Agricultural Products – Adding Value

Processing means **changing raw farm produce into a more useful, durable, or valuable form**.

- **Why Process?**
 - a. **Prevents Spoilage:** Fresh tomatoes rot in 3 days. Tomato paste lasts months.

- b. **Adds Value:** 1kg of fresh cassava sells for less than 1kg of garri. The processing (peeling, grating, frying) adds value and profit.
- c. **Creates Variety:** Maize can become corn flour, pap (ogi), popcorn, or corn flakes.
- **Common Nigerian Examples:**
 - Cassava → **Garri**, Fufu, Starch
 - Oil Palm Fruit → **Palm Oil**
 - Milk → **Yoghurt**, Cheese
 - Groundnut → **Groundnut Oil**, Kulikuli



3. Record Keeping – The Farmer's Diary

Record keeping is the skill of **writing down important information** about the farm.

- **What to Record:**
 - **Money In/Out:** Cost of seeds, feed, medicine. Income from sales.
 - **Farm Activities:** Date of planting, weeding, spraying, harvesting.
 - **Animal Records:** Number of eggs laid daily, birth of new animals, health treatments given.

- **Importance:** Records tell the farmer if the business is making a **profit or a loss**. They help in planning for the next season and solving problems.

4. Marketing Farm Produce – Selling Smartly

Marketing is all the activities involved in **getting your produce from the farm to the customer who will buy it**.

- **Skills Involved:**
 - **Pricing:** Knowing how to set a fair price that covers your costs and gives a profit, but is not too high to drive buyers away.
 - **Presentation:** Cleaning, sorting, and packaging produce attractively. No one wants to buy dirty or bruised tomatoes.
 - **Finding Buyers:** Knowing where to sell—local market, roadside stand, to a wholesaler, or even to a processing company.

EVALUATION

1. What is one key difference between feeding a day-old chick and a mature hen?
2. Explain one reason why processing cassava into garri is better than selling raw cassava.
3. How can a farmer use his records to decide what to plant next year?
4. If a farmer spends ₦5,000 on seeds and fertilizer and sells his harvest for ₦8,000, what is his profit? How did record keeping help him know this?
5. Name two things a farmer can do to make his vegetables more attractive to buyers in the market.
6. True or False: Marketing is only about shouting the loudest in the market.

ASSIGNMENT

1. **Survey:** List five processed food products you ate this week (e.g., Bread, Garri, Yoghurt). Next to each, write the raw agricultural product it came from (e.g., Wheat, Cassava, Milk).

2. **Practice:** Create a one-day record for an imaginary chicken farmer. Write down: Date, Number of eggs collected, Amount of feed given, and one note (e.g., "One hen looks sick").
3. **Role-play:** With a partner, act out a buying and selling scene at a market. The seller should try to explain why their produce (e.g., clean, fresh carrots) is worth the price.



WEEK 5 & 6: MIDTERM EXAMINATION AND BREAK



WEEK 7: FOOD CLASSIFICATION AND FUNCTIONS I

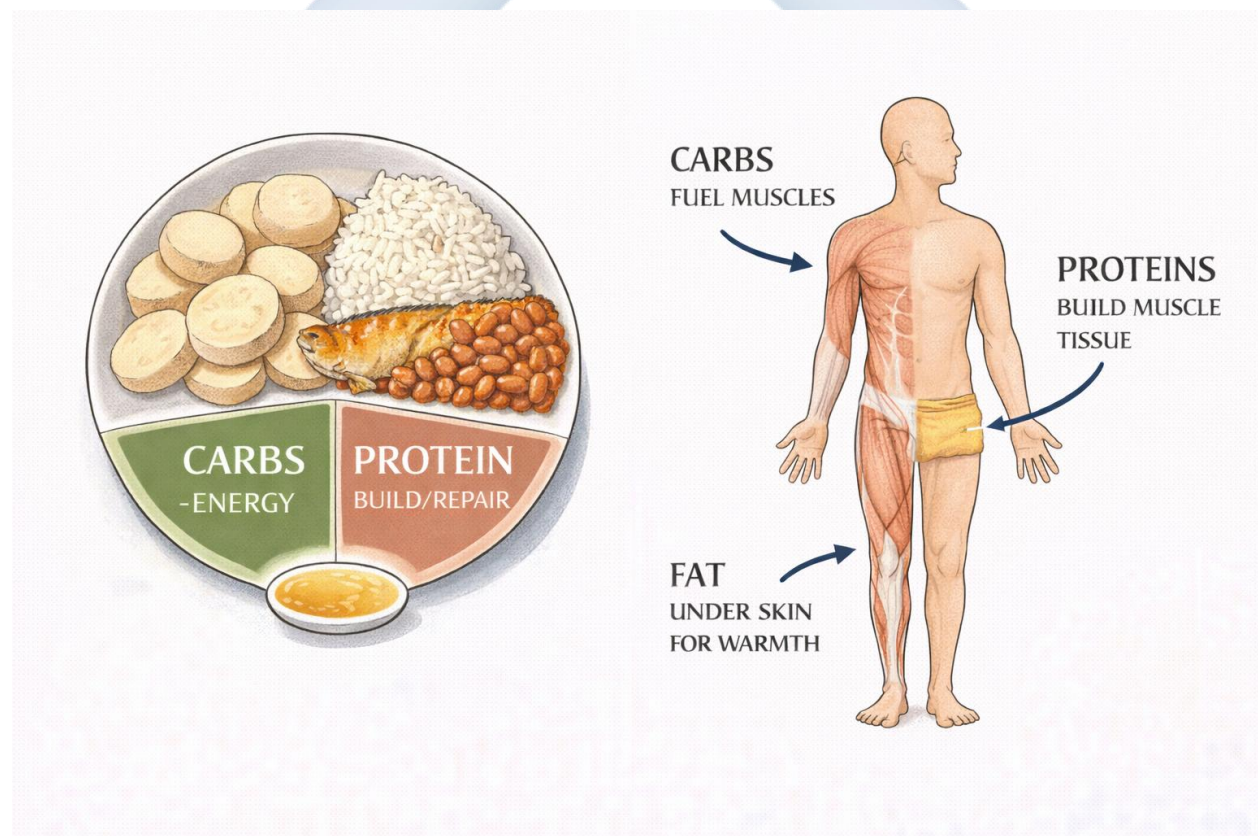
OBJECTIVES By the end of the lesson, pupils should be able to:

1. List the three major classes of food (Macronutrients).
2. Mention three sources of carbohydrates.
3. State the function of protein in the body.
4. Identify sources of fats and oils.

CONTENT

1. Classification of Food – The Body's Fuel System

The food we eat can be grouped into **classes** based on the main job they do for our body. Think of your body as a car. Different foods provide different types of "fuel" and "materials" for the journey of life.



2. Carbohydrates – The Energy Givers (Body Fuel)

- **Main Function:** To provide **ENERGY** for all activities—thinking, breathing, running, playing football. They are the body's **preferred source of quick fuel**.
- **Chemical Make-up:** Made up of sugars and starches.
- **Common Nigerian Sources:**
 - **Staples:** Yam, Rice, Garri/Eba (from cassava), Maize/Corn, Wheat (in bread).
 - **Others:** Potatoes, Plantain, Cocoyam, Oats.
- **Simple Analogy:** Carbohydrates are like the **petrol** in a car's tank.

3. Proteins – The Body Builders and Repairers (Body Building Blocks)

- **Main Function:**
 - a. **Growth:** They are the main building materials for growing new tissues—making you taller, building muscles.
 - b. **Repair:** They repair worn-out or damaged tissues—healing cuts, mending bruises, replacing old cells.
- **Chemical Make-up:** Made up of smaller units called **amino acids**.
- **Common Nigerian Sources:**
 - **Animal Sources (Complete Proteins):** Meat (beef, chicken, goat), Fish, Eggs, Milk, Cheese.
 - **Plant Sources:** Beans (all types: brown, white, black-eyed), Soya beans, Groundnuts, Melon seeds (Egusi).
- **Simple Analogy:** Proteins are like the **bricks and cement** used to build and repair a house.

4. Fats and Oils – The Energy Bank and Insulators

- **Main Function:**
 - a. **Stored Energy:** They provide a **concentrated, long-lasting store of energy** to be used when carbohydrates run out.
 - b. **Insulation & Protection:** They form a layer under the skin that **keeps the body warm**. They also cushion vital organs (like the heart and kidneys) like soft pillows.

- **Sources:**

- **Oils (Liquid at room temp):** Palm oil, Vegetable oil, Groundnut oil, Olive oil.
- **Fats (Solid at room temp):** Butter, Margarine, Fat on meat, Avocado, Coconut.

EVALUATION

1. Which class of food would a runner eat a lot of before a race to get quick energy?
2. A child has a deep cut. Which class of food will be most important for healing the wound?
3. Name three common Nigerian foods that are rich in carbohydrates.
4. Why are beans and groundnuts important for children who don't eat meat often?
5. What are the two main jobs of fats in the body?
6. True or False: Eating only pounded yam (carbohydrate) for every meal gives your body everything it needs.

ASSIGNMENT

1. **Food Diary:** Write down everything you ate for dinner yesterday. Next to each item, write its main food class (C for Carbohydrate, P for Protein, F for Fat).
2. **Drawing:** Draw a "Food Family." Create characters for "Mr. Energy Carb," "Builder Protein," and "Warmth-Fat." Show what they do for the body.
3. **Investigation:** Look at the ingredients list on a pack of biscuits or a cooking oil bottle at home. Can you spot any words related to fats/oils or carbohydrates? Write one down.

WEEK 8: FOOD CLASSIFICATION AND FUNCTIONS II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Explain the function of Vitamins and Minerals.
2. List sources of Vitamins.
3. Define a Balanced Diet.
4. State the importance of Water and Roughage.

CONTENT

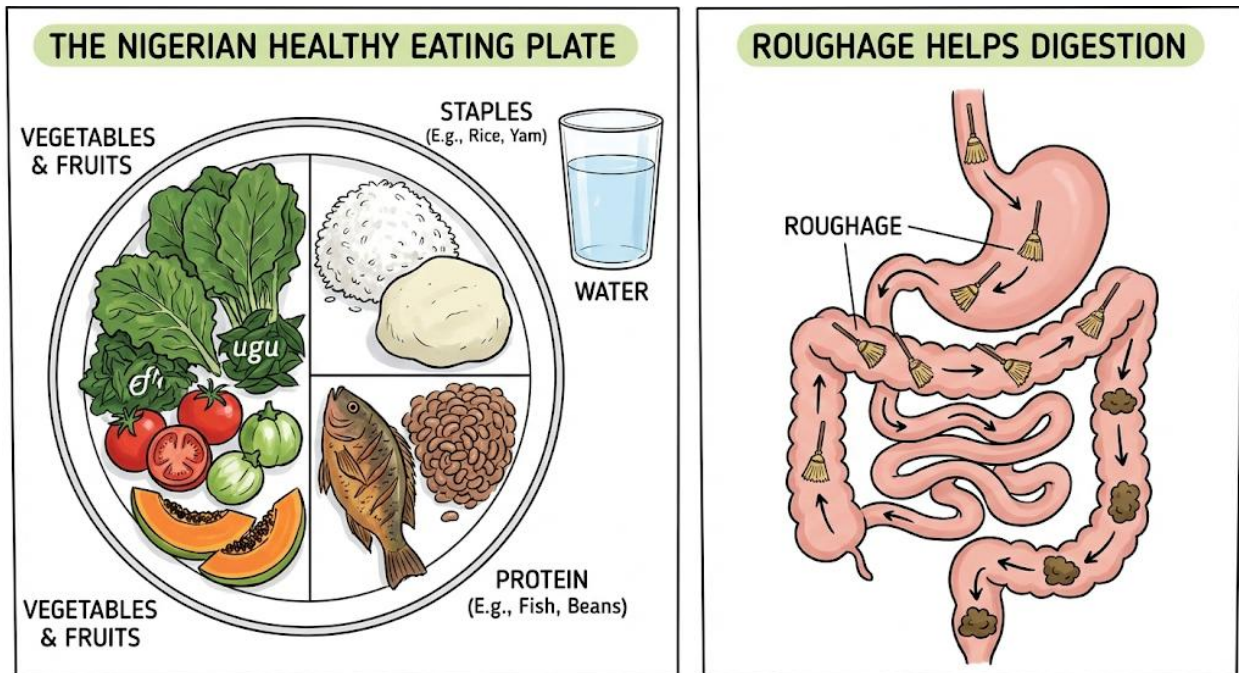
1. Vitamins and Minerals – The Protective Regulators (Body's Spark Plugs and Armour)

- **Main Function:** They do not provide energy or building materials. Instead, they **protect the body from disease** and **help regulate** all the body's processes to keep it running smoothly.
 - **Vitamins:** Help with vision (Vitamin A), healing (Vitamin C), strong bones (Vitamin D), and blood clotting (Vitamin K).
 - **Minerals:** Build strong bones and teeth (Calcium, Phosphorus), make blood (Iron), help nerves and muscles work (Potassium, Sodium).
- **Sources: Fresh fruits and vegetables** are the powerhouses!
 - **Fruits:** Oranges, Pineapples, Mangoes, Watermelon, Pawpaw, Guava, Bananas.
 - **Vegetables:** Ugwu (fluted pumpkin), Efo (spinach), Okra, Carrots, Tomatoes, Onions, Green peppers.
 - **Others:** Milk (Calcium), Liver (Iron), Sea salt (Iodine).

2. Water – The Essential Transport System

- **Function:** Water is **the most critical** nutrient. It is not a food class but is essential for life. It:
 - Carries nutrients and oxygen to all cells.
 - Removes waste from the body (sweat, urine).
 - Regulates body temperature.
 - Is a major part of blood, saliva, and digestive juices.

- **Need:** We need to drink **clean water** regularly throughout the day, especially when it's hot or when we are active.



3. Roughage (Dietary Fibre) – The Internal Cleaner

- **Function:** Roughage is the **indigestible part** of plant foods. It acts like a **broom** in our digestive system.
 - It adds bulk to waste (stool), helping it move easily through the intestines and preventing **constipation** (hard, difficult stool).
 - It helps you feel full.
- **Sources:** The **skin, seeds, and pulp** of fruits and vegetables, whole grains (like unpolished rice), beans.

4. Balanced Diet – The Complete Meal

- **Definition:** A **balanced diet** is a meal that contains **all the six classes of food in the right proportions**: Carbohydrates, Proteins, Fats & Oils, Vitamins, Minerals, and Water.
- **Why it's Important:** Eating only one class (like only rice and bread) leads to **malnutrition**—either deficiency diseases (like scurvy from lack of Vitamin C) or obesity.

A balanced diet ensures the body gets everything it needs to grow, be active, and stay healthy.

- **Example of a Balanced Nigerian Meal:**

- **Jollof Rice** (Carbohydrate + Vitamins from tomatoes)
- **Grilled Chicken** (Protein + some Fat)
- **Steamed Cabbage/Carrots** (Vitamins, Minerals, Roughage)
- **A glass of water**
- **A slice of orange for dessert** (Vitamins, Roughage)

EVALUATION

1. Why are vitamins and minerals called "protective foods"?
2. Name two fruits that are excellent sources of Vitamin C.
3. What is the main problem that roughage helps to prevent in the body?
4. Describe one vital job that water does inside your body.
5. Look at this meal: White Rice + Stew (with only oil and pepper) + Coke. Is it balanced? Explain what is missing.
6. True or False: Drinking Fanta or Coca-Cola is the same as drinking water for hydration.

ASSIGNMENT

1. **Creative:** Plan a balanced lunch menu for your family. List the dishes and next to each, write the main food class it provides.
2. **Practical:** Eat a fruit (like an orange or mango). Keep the peel or seeds. Paste them in your book and write: "This is a source of roughage."
3. **Experiment:** Observe for one day how many times you drink water. Try to increase it by one extra cup tomorrow. Write a sentence about how you feel.

WEEK 9: BASIC PRINCIPLES IN HOME ECONOMICS I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List four methods of cooking.
2. Differentiate between boiling and frying.
3. Mention two kitchen safety rules.
4. Explain kitchen hygiene.

CONTENT

1. Principles of Cooking – Why and How We Cook We cook food to:

- Make it **safe** by killing harmful germs and parasites.
- Make it **easier to digest** and absorb nutrients.
- Improve its **taste, texture, and appearance**.
- Give **variety** to our meals.

2. Common Cooking Methods

- **Boiling:**
 - **Process:** Cooking food in **hot water** at 100°C.
 - **Examples:** Boiling yam, rice, eggs, potatoes.
 - **Key Point:** Nutrients can leak into the water (which is why we make stews with the water from boiled meat).
- **Frying:**
 - **Process:** Cooking food in **hot oil**.
 - **Types:** **Deep frying** (food submerged in oil: akara, puff-puff) and **Shallow frying** (food cooked in a little oil: fried eggs, plantain).
 - **Note:** Too much fried food is unhealthy.
- **Roasting/Grilling:**
 - **Process:** Cooking food by **direct dry heat** from fire, coal, or an oven grill.
 - **Examples:** Roasted corn (bole), suya, roasted fish, grilled chicken.

- **Steaming:**

- **Process:** Cooking food by the **steam** from boiling water, without the food touching the water.
- **Examples:** Moin-moin, okpa, custard. This method preserves more nutrients.



3. Safety in Food Preparation – The Kitchen is Not a Playground

- **Handling Heat:** Use **dry pot holders** or folded cloth to handle hot pots. A wet cloth transmits heat quickly and will burn your hand.
- **Knife Safety:** Always cut **away from your body**. Keep fingers curled in (the "claw grip") when holding food to cut.
- **Pot Handles:** Turn pot handles **inward** on the stove, not over the edge where they can be knocked over.
- **Fire Safety:** Know where the fire extinguisher is. If a small grease fire starts, **smother it** by covering the pan with a lid and turn off the heat. **Never pour water on it.**

4. Hygiene in the Kitchen – Clean Food is Safe Food

Kitchen hygiene prevents food poisoning.

- **Personal Hygiene:** Wash hands with soap before and during cooking, especially after touching raw meat. Cover hair.
- **Food Hygiene:** Wash all fruits and vegetables thoroughly. Keep raw meat separate from ready-to-eat food.
- **Equipment Hygiene:** Use clean knives and cutting boards. Wash dishes with hot, soapy water.
- **Work Area Hygiene:** Keep surfaces clean and dry. Dispose of waste properly.

EVALUATION

1. What is one reason we cook food that is not about taste?
2. Which cooking method is used to make moin-moin, and how does it work?
3. Why is it dangerous to use a wet cloth to pick up a hot pot?
4. What should you do immediately if a small amount of oil in a pan catches fire?
5. Why is it important to use different cutting boards for raw meat and for vegetables?
6. True or False: Roasting and frying are the same because they both use heat.

ASSIGNMENT

1. **Practical Observation:** Watch someone cook at home. Identify and write down which cooking method(s) they used (boiling, frying, etc.).
2. **Poster Design:** Create a "Kitchen Safety Rule" poster with two important rules. Use drawings and clear words.
3. **Hygiene Audit:** Check your kitchen at home (with permission). Write down one thing that is very hygienic (e.g., soap at the sink) and one thing that could be improved (e.g., a wet cloth left on the counter).

WEEK 10: BASIC PRINCIPLES IN HOME ECONOMICS II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define sewing.
2. List three items used for sewing.
3. Mention two types of stitches.
4. State one principle of home cleaning.

CONTENT

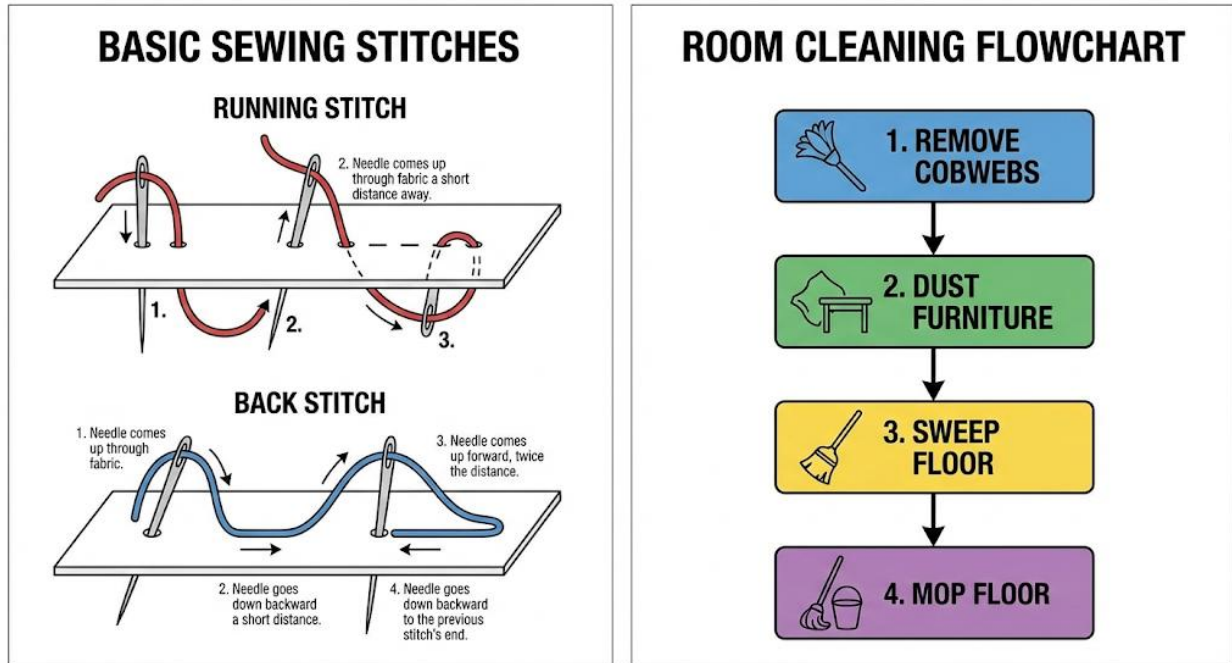
1. Principles of Sewing – Joining Fabrics

- **Sewing** is the craft of **fastening or attaching objects** using stitches made with a **needle and thread**. It is used to make, mend, or decorate clothes and household items.
- **Basic Sewing Tools:**
 - **Needle:** A thin, pointed metal tool with an **eye** (hole) to hold the thread.
 - **Thread:** A long, thin strand of cotton, polyester, or silk used to form stitches.
 - **Scissors:** Sharp blades for cutting fabric and thread. **Fabric scissors** should only be used for fabric to keep them sharp.
 - **Tape Measure:** A flexible ruler for taking body and fabric measurements.
 - **Pins and Pincushion:** Pins hold fabric pieces together before sewing. The pincushion stores them safely.

2. Basic Hand Stitches

- **Running Stitch:**
 - **How:** The simplest stitch. The needle goes **in and out** of the fabric in a straight line, creating a dashed line.
 - **Use:** For simple seams, gathering fabric, or temporary basting.
- **Back Stitch:**
 - **How:** A very strong stitch. Start with a small stitch forward. For the next stitch, go back to the end of the previous one and come up a stitch length ahead. This creates an unbroken line.

- **Use:** For strong seams that need to hold under stress, like a bag strap.
- **Hemming Stitch (Slip Stitch):**
 - **How:** A nearly invisible stitch used to fold and secure the edge of fabric (a hem).
 - **Use:** For finishing the bottom of skirts, trousers, or sleeves.



3. Principles of Home Cleaning – Order and Health

A clean home is healthy, pleasant, and safe.

- **Systematic Approach:** Clean from **top to bottom** (dust ceilings/cobwebs first) and **from dry to wet** (sweep before you mop).
- **Sweeping:** Use a broom to sweep dirt from corners to the center of the room, then collect it with a dustpan.
- **Dusting:** Use a dry or slightly damp cloth to wipe surfaces. Dust **before** sweeping so that dust from furniture falls to the floor to be swept away.
- **Organisation:** "A place for everything and everything in its place." This makes cleaning easier and finds things faster.

4. Home Management Principles

- **Resource Management:** This means using things wisely to avoid waste. Turn off lights and fans when leaving a room. Fix leaking taps to save water.
- **Preventive Maintenance:** Repair a small tear in a shirt immediately before it becomes a big hole. Tighten a loose screw on a chair before it gets lost.

EVALUATION

1. What is the main purpose of a tape measure in sewing?
2. Which hand stitch would you use to securely mend a torn school bag strap? Why?
3. Why should you dust shelves before sweeping the floor?
4. What does the saying "A stitch in time saves nine" mean in terms of home management?
5. How does turning off an unused light bulb relate to home economics?
6. True or False: It is okay to use fabric scissors to cut paper if you can't find your paper scissors.

ASSIGNMENT

1. **Practical Skill:** On a piece of old cloth or thick paper, practice making a line of running stitches. Paste your sample in your book.
2. **List & Draw:** List three tools used for cleaning your home. Choose one and draw it, labeling its parts (e.g., a broom: handle, bristles).
3. **Observation:** In your home this week, find one example of "preventive maintenance" (something small that was fixed). Write a sentence about it.

WEEK 11: REVISION

CONTENT

1. Comprehensive Review of Term 2 Topics

Let's recap the major themes of this term:

- **Tool Maintenance (Weeks 1-2):** Cleaning, drying, sharpening, oiling, and proper storage of farm tools to ensure longevity, efficiency, and safety.
- **Vocational Skills (Weeks 3-4):** The practical, hands-on skills in agriculture: crop production (land prep, planting, weeding, harvesting), animal rearing, processing, record keeping, and marketing.
- **Food and Nutrition (Weeks 7-8):** The six classes of food (Carbohydrates, Proteins, Fats & Oils, Vitamins, Minerals, Water & Roughage) and the importance of a Balanced Diet for health and growth.
- **Home Economics Basics (Weeks 9-10):** Cooking methods (boiling, frying, roasting, steaming), kitchen safety and hygiene, basic sewing, and principles of home cleaning and management.

2. Interactive Revision Activities

- **"Sort It Out" Game:** The teacher holds up cards with names of tools, food items, or skills. Pupils must shout out the correct category (e.g., "Cutlass!" -> "Farm Tool for Clearing"; "Orange!" -> "Vitamin C Source").
- **"Fix the Problem" Scenarios:** Pupils work in small groups to solve a problem.
 - *Example 1:* "A farmer's hoe is rusty and the handle is loose. What steps should he take to fix it?"
 - *Example 2:* "A meal consists of only eba and egusi soup with no meat or fish. What food class is missing, and what could be added to balance it?"
- **Picture-Based Quiz:** The teacher shows pictures (a jar test, a sewing needle, a knapsack sprayer) and asks pupils to explain what is happening or identify the item and its use.

EVALUATION (*This is a formative, revision-focused evaluation to consolidate learning.*)

1. Arrange these steps in the correct order for cleaning a tool after use: (Dry, Scrape, Wash).
2. Which vocational skill helps a farmer know if he is making money or losing money?
3. If someone is feeling weak and tired all the time, which class of food might they not be eating enough of?
4. What is the key safety reason for turning pot handles inward on a stove?
5. Name one source of roughage and explain what it does in the body.
6. Why is it important to learn both agricultural and home economics skills? (They help us in producing food and managing the home effectively).

ASSIGNMENT

1. **Study Guide:** Create a one-page summary poster for Term 2. Divide it into four boxes (Maintenance, Voc Skills, Food, Home Ec) and put 2 key words or drawings in each.
2. **Teach Back:** Explain one important thing you learned this term to an older person at home (parent, aunt, uncle). Ask them if they knew it. Write down their reaction.
3. **Preparation:** Look through your notebook and identify one topic you found a bit challenging. Write down one question about it to ask during the next revision class.

PRIMARY 5 PREVOCATIONAL STUDIES SCHEME OF WORK

THIRD TERM

Week	Topic	Content
1	Home Economics Tools and Equipment I	Kitchen equipment - pots, pans, knives, cutting boards, measuring cups; Sewing equipment - needles, thread, scissors, measuring tape
2	Home Economics Tools and Equipment II	Cleaning equipment - brooms, mops, vacuum cleaners, dusters; Laundry equipment - washing machine, iron, drying rack; Uses of each equipment
3	Safety in Handling Home Economics Tools I	Kitchen safety - knife handling, hot surfaces, electrical appliances; Prevention of cuts and burns; Fire safety
4	Safety in Handling Home Economics Tools II	Sewing safety - needle and scissors safety, iron safety; Electrical safety; Chemical safety - detergents and cleaning agents
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Maintenance of Home Economics Tools	Cleaning kitchen equipment; Sharpening knives safely; Caring for sewing equipment; Storing tools properly; Regular checks
8	Personal Grooming and Use of Basic Cosmetics I	Advanced personal grooming; Skin care basics; Hair care; Nail care; Age-appropriate grooming
9	Personal Grooming and Use of Basic Cosmetics II	Safe use of simple cosmetics; Natural beauty products; Personal hygiene products; Making informed choices; Self-esteem and appearance
10	Introduction to Entrepreneurship	Meaning of entrepreneurship; Young entrepreneurs; Turning skills into business; Agriculture and Home Economics businesses; Creativity and innovation
11	REVISION	Review of all topics covered; Practice exercises; Question and answer sessions

12	EXAMINATION	End of term assessment
13	CLOSING	Report card distribution; Vacation assignment



PRIMARY 5 PRE-VOCATIONAL STUDIES LESSON NOTES

THIRD TERM

WEEK 1: HOME ECONOMICS TOOLS AND EQUIPMENT I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List three examples of kitchen equipment.
2. Identify measuring tools used in the kitchen.
3. Mention four items found in a sewing kit.
4. State the function of a tape measure.

CONTENT

1. Kitchen Equipment – Tools for Food Preparation

Kitchen equipment includes all the **tools, utensils, and appliances** we use to prepare, cook, and serve food. Using the right tool makes cooking easier, safer, and more enjoyable.

- **Pots and Pans:** These are containers for cooking on a stove or fire.
 - **Pot:** Deep with two handles, for boiling soups, stews, rice (e.g., stainless steel pot, traditional iron pot "*orokoro*").
 - **Frying Pan (Saucepan):** Shallow with one long handle, for frying eggs, plantains, meat.
- **Knives:** Sharp tools for cutting.
 - **Different Sizes:** A large **kitchen knife** (for meat/vegetables), a smaller **paring knife** (for peeling fruits), and a **bread knife** (with a serrated edge).
- **Cutting Board:** A flat board made of wood or plastic. It protects your table or countertop from knife cuts and provides a safe, stable surface for chopping.
- **Mixing Bowls:** Deep, round bowls in various sizes used for mixing ingredients together, like whisking eggs, mixing flour for cakes, or tossing a salad.

2. Measuring Tools in the Kitchen – For Perfect Results

Cooking is like a science. To make food taste the same every time and to follow recipes, we need to measure accurately.

- **Measuring Cups:** Used for measuring **dry ingredients** like flour, sugar, rice, and beans. They come in sets ($\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$, 1 cup).
- **Measuring Spoons:** Used for small amounts of dry or wet ingredients like **salt, baking powder, pepper, or vanilla flavour**. Sets include spoon sizes like $\frac{1}{4}$ tsp, $\frac{1}{2}$ tsp, 1 tsp, 1 tbsp.
- **Liquid Measuring Jug:** Usually made of clear glass or plastic with markings (millilitres -ml- and cups) on the side. Used to measure **water, milk, oil, or stock**.



3. Sewing Equipment – Tools for Making and Mending

A basic sewing kit contains tools for simple repairs and projects.

- **Needles:** Thin, pointed metal tools with a hole at one end called the **eye**. Different needles exist for different fabrics (sharps for general sewing, betweens for detailed work).

- **Thread:** Long, thin strands of cotton, polyester, or silk that are passed through the needle's eye to make stitches. Thread colour should match the fabric.
- **Scissors:**
 - **Fabric Scissors:** Large, sharp scissors used *only* for cutting cloth to keep them sharp.
 - **Small Embroidery Scissors:** For snipping thread.
- **Pins and Pincushion:** Straight pins hold fabric pieces together temporarily before sewing. They are stuck into a **pincushion** (a soft pad) for safe and easy access.

4. The Tape Measure – The Tailor's Rule

- **Description:** A long, thin, flexible strip of cloth, plastic, or fibreglass. It is marked with measurements in both **centimetres (cm)** and **inches (")**** on opposite sides. It rolls up into a small case.
- **Functions:**
 - a. **Taking Body Measurements:** To make or buy clothes that fit, you measure parts of the body (e.g., waist, chest, arm length).
 - b. **Measuring Fabric:** To know how much cloth is needed for a project.
 - c. **General Measuring:** Around the home for other tasks.

EVALUATION

1. Why is it important to use a cutting board instead of cutting directly on a table?
2. If a recipe asks for "one cup of flour," which specific tool would you use to measure it?
3. What is the main difference between *fabric scissors* and *paper scissors*? Why is this important?
4. What are pins used for in sewing, and what tool keeps them safe when not in use?
5. A tailor needs to know how long your arm is to make a shirt sleeve. Which tool will he use?
6. True or False: You can use the same cup you drink from to accurately measure one cup of rice for cooking.

ASSIGNMENT

1. **Drawing:** Draw and label three different kitchen tools you have at home. Write one use for each.
2. **Practical Investigation:** Find a tape measure at home (ask for permission). Measure the length of your notebook in centimetres (cm). Write down the measurement.
3. **List & Sort:** Look in your kitchen. Make a list of two tools used for **cutting** and two tools used for **measuring**.



WEEK 2: HOME ECONOMICS TOOLS AND EQUIPMENT II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List three tools used for cleaning the home.
2. Explain the function of a vacuum cleaner.
3. Mention three items used for laundry.
4. State the use of a drying rack.

CONTENT

1. Cleaning Equipment – Tools for a Healthy Home

Keeping our home clean prevents illness and makes it pleasant to live in.

- **Brooms:**
 - **Soft Broom (Brush):** For sweeping smooth floors inside the house (tiled, cemented). It gathers fine dust.
 - **Hard Broom (Palm frond/broomstick):** For sweeping rough surfaces, compounds, and removing cobwebs.
- **Dustpan and Brush:** A small pan and a short-handled brush used together to collect swept-up dirt from the floor and transfer it to the bin.
- **Mop and Bucket:** Used for washing floors. The **mop** (with a cloth or sponge head) is dipped in soapy water in the **bucket**, wrung out, and then used to clean the floor.
- **Dusters:** Soft cloths (like old T-shirts), feather dusters, or synthetic dusters used to **wipe away dust** from furniture, shelves, fans, and electronics.



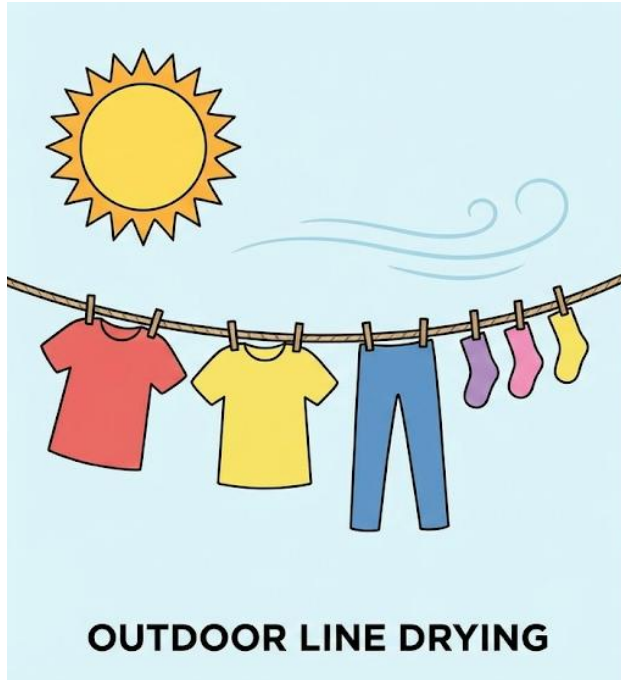
2. The Vacuum Cleaner – A Power Sweeper

- **What it is:** An **electrical appliance** that cleans floors by suction. It has a motor that creates a vacuum (suction), a bag or container for dirt, and different attachments.
- **How it works:** It **sucks in** dust, dirt, crumbs, and even tiny insects from floors, carpets, rugs, and upholstery through a hose or nozzle.
- **Advantages:** It is very **effective and fast**, especially on carpets where brooms just push dust around. It also helps people with allergies by removing dust from the air.

3. Laundry Equipment – Tools for Washing Clothes

Laundry is the process of washing, drying, and ironing clothes.

- **Washing Machine:** An electrical appliance that automates washing. You add clothes, detergent, and water, and it **agitates** (moves them around) to clean them, then **spins** to remove most of the water.
- **Iron:** An electrical appliance with a flat, heated metal plate (soleplate). It uses **heat and pressure** to remove wrinkles and creases from fabric, making clothes look neat and smart.
- **Laundry Basket/Hamper:** A basket, bag, or bin where dirty clothes are collected before wash day. This keeps dirty clothes from lying around.



4. The Drying Rack (Clothes Horse)

- **Description:** A portable, foldable frame made of metal, wood, or plastic with many horizontal bars.
- **Function:** It is used to **air-dry** wet clothes indoors or in a shaded outdoor area. Clothes are spread over the bars.
- **Why use it?**
 - When it is raining and you cannot use an outdoor clothesline.
 - For delicate clothes that shouldn't be in strong sunlight.
 - In homes without a garden or balcony.

EVALUATION

1. When would you choose to use a hard broom instead of a soft broom?
2. What is the main job of a dustpan?
3. How is a vacuum cleaner better than a broom for cleaning a carpet?
4. What are the two main things (heat and ?) that an iron uses to smooth clothes?
5. Where should you put your dirty school uniform when you take it off?
6. Give two reasons why a family might use a drying rack instead of an outdoor line.

ASSIGNMENT

1. **Drawing:** Draw two tools used for cleaning floors in your home. Label them.
2. **Practical:** Help with a cleaning task at home (e.g., sweeping a room, dusting a table).
Write down which tool you used and one safety tip you followed (e.g., "I swept gently so dust didn't fly up").
3. **Observation:** Look at how clothes are dried in your home. Do you use a line, rack, or both? Draw a simple picture of the method.



WEEK 3: SAFETY IN HANDLING HOME ECONOMICS TOOLS I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Mention two rules for handling knives safely.
2. Explain how to prevent burns from hot surfaces.
3. State one rule for electrical safety in the kitchen.
4. Describe what to do in case of a small kitchen fire.

CONTENT

1. Kitchen Safety: Knife Handling – Your Hands are Precious

Knives are among the most dangerous tools in the home if used carelessly.

- **Rule 1: The Claw Grip.** When holding food to cut, **curl your fingertips inward** like a claw, using your knuckles to guide the knife. This keeps your fingertips safe.
- **Rule 2: Cut Away, Never Towards.** Always cut **away from your body** and from the hand holding the food.
- **Rule 3: Safe Carrying and Storage.** When walking with a knife, hold it **pointing down** at your side. Store knives in a **knife block** or on a magnetic strip, **not** loose in a drawer where you can grab the blade.
- **Rule 4: Never Catch a Falling Knife.** If you drop a knife, **jump back!** Let it fall. Never try to catch it.

2. Preventing Burns and Scalds

Burns from hot pots, oil, steam, or water are common kitchen injuries.

- **Use Dry Pot Holders/Oven Mitts:** Always use a **thick, dry cloth or specially made mitt** to handle hot pots, pans, or trays. A **wet cloth will steam and burn you instantly** because heat travels quickly through water.
- **Turn Handles Inward:** On the stove, turn pot handles **towards the centre, not over the edge**. This prevents you or a child from accidentally bumping the handle and tipping hot food over.

- **Open Lids Away from You:** When opening a pot of boiling food (like rice), tilt the lid away from your face to let hot steam escape safely.



3. Electrical Appliance Safety

The kitchen has many electrical tools. Water and electricity are a deadly mix.

- **Dry Hands:** Never touch plugs, switches, or appliances with **wet hands**.
- **Check Cords:** Don't use appliances with **frayed, cut, or damaged cords**.
- **Unplug Safely:** Pull the **plug itself**, not the cord, to disconnect an appliance.
- **Keep Away from Water:** Don't place electric kettles, blenders, or toasters near the sink where water can splash.

4. Fire Safety – Don't Panic, Act!

- **Prevention:** Don't leave cooking food unattended on the stove.
- **If a Small Pan Fire Starts:**
 - a. **STOP.** Don't panic.
 - b. **TURN OFF** the heat source (stove).

- c. **COVER** the pan with a metal lid or a damp cloth to **smother** the flames by cutting off oxygen. **DO NOT USE WATER**, especially on an oil/grease fire, as it will cause a dangerous explosion of hot oil.
- d. **LEAVE** the pan covered until it is completely cool.
- e. **CALL** for an adult immediately.

EVALUATION

1. Why is the "claw grip" important when using a knife?
2. What is the safest way to carry a knife from the sink to the cutting board?
3. Explain why a wet cloth is dangerous for picking up a hot pot.
4. If a plug is near the sink, why is it unsafe to touch it after washing your hands if they are still wet?
5. What is the **first** thing you should do if oil in a frying pan catches fire?
6. True or False: Pouring water on a small fire is always the right thing to do.

ASSIGNMENT

1. **Practical Demonstration:** With a **butter knife** (not sharp) and a banana or soft vegetable, practice the "claw grip" in front of a family member. Explain to them why it's safe.
2. **Safety Poster:** Design a "Kitchen Fire Safety" poster. Include the rule: "**Turn Off, Cover, Call for Help**" and a warning: "**Never Use Water on Grease Fire!**"
3. **Home Survey:** Check your kitchen at home (with an adult). Find one electrical appliance (like a kettle). Is its cord in good condition? Is it kept away from water? Write your observation.

WEEK 4: SAFETY IN HANDLING HOME ECONOMICS TOOLS II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Mention safety rules for using needles and scissors.
2. Explain how to use an iron safely.
3. Define chemical safety.
4. State one rule for storing cleaning agents.

CONTENT

1. Sewing Safety – Sharp Points and Edges

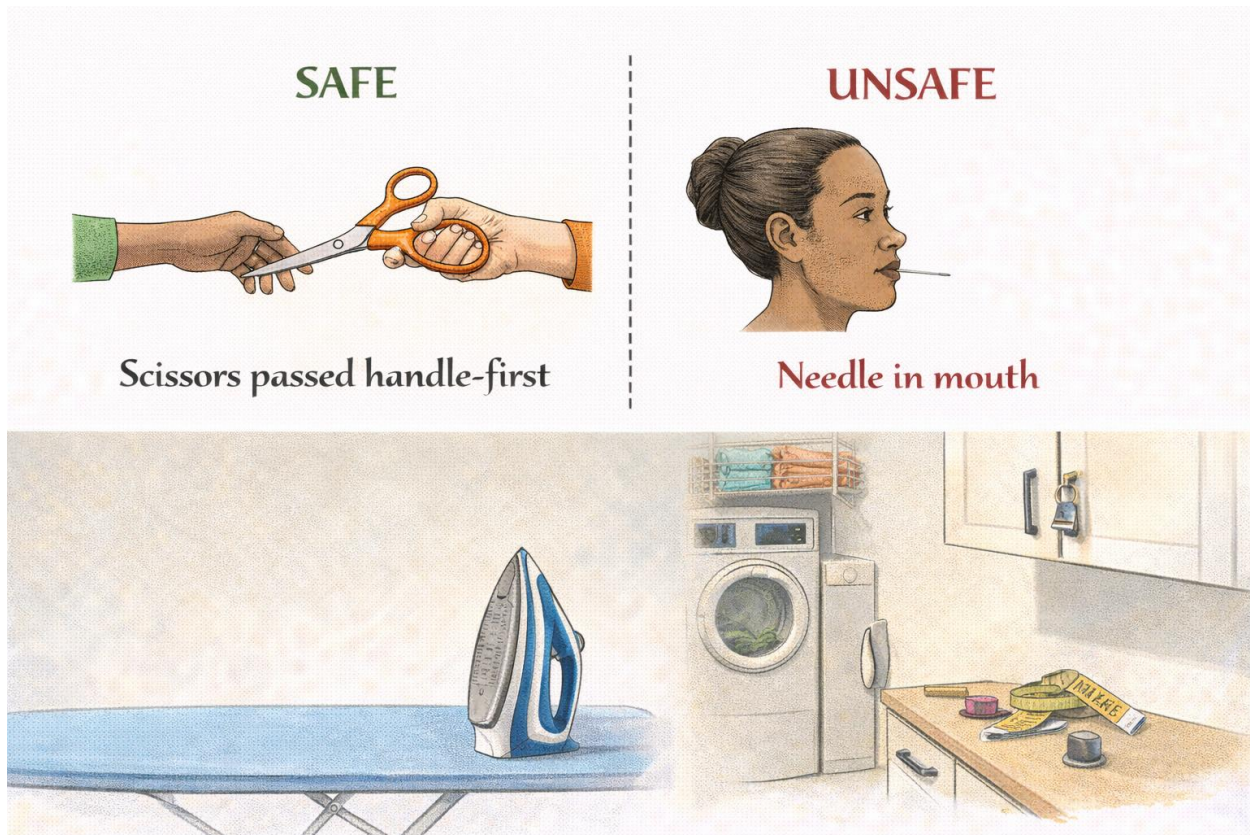
- **Needle and Pin Safety:**
 - **Never put pins or needles in your mouth.** You could accidentally swallow or inhale them.
 - Store them in a **pincushion or a labelled box.**
 - When threading a needle, point it **away from your eyes.**
- **Scissor Safety:**
 - Always pass scissors to someone **handle-first**, with the blades closed.
 - **Never run** while holding scissors.
 - When not in use, close the blades and store them safely.

2. Iron Safety – Respect the Heat

An iron is very hot and can cause bad burns or start a fire.

- **Cord Safety:** Check that the cord is not damaged before plugging it in. Don't let the cord touch the hot soleplate.
- **Stable Ironing Board:** Use a proper ironing board that is stable and at the right height. Don't iron on beds or floors covered with flammable materials.
- **Safe Resting:** When taking a break, **stand the iron upright on its heel** on the ironing board. **Never** leave it lying flat on the board or fabric.

- **Unplug and Cool:** Always **unplug the iron immediately** after use. Let it cool down completely in a safe place before storing it.



3. Electrical Safety (General for the Home)

- **Don't Overload Sockets:** Plugging too many appliances (like TV, decoder, fan, phone charger) into one socket or extension cord can cause overheating and fire.
- **Switch Off and Unplug:** When appliances are not in use (like a hair dryer or pressing iron), switch them off at the socket and unplug them.

4. Chemical Safety – Cleaners are Not Toys

Many home cleaning products are **chemicals** that can be poisonous, corrosive, or flammable.

- **Definition of Chemical Safety:** It means **using, storing, and disposing of household chemicals in a way that prevents harm to people, pets, and the environment.**
- **Golden Rules:**

- **Never Mix Chemicals:** Mixing cleaners (e.g., bleach and toilet cleaner) can create **poisonous gas**.
- **Read Labels:** Always follow the instructions and warnings on the container.
- **Safe Storage:** Keep all chemicals (bleach, detergent, insecticides) in their **original containers** with clear labels. Store them **up high, in a locked cabinet, away from children and food**.
- **Never Transfer:** **Never** put a chemical (like kerosene or pesticide) into a **drink bottle, cup, or bowl**. Someone might mistake it for water and drink it, which can be fatal.

EVALUATION

1. What is the danger of putting a sewing pin in your mouth?
2. How should you correctly pass a pair of scissors to your friend?
3. Why must you stand a hot iron upright on its heel instead of laying it flat?
4. What dangerous thing can happen if you mix bleach and toilet cleaner?
5. Why is it extremely dangerous to store petrol or "otapiapia" (insecticide) in an empty soft drink bottle?
6. True or False: It's okay to plug a fridge, microwave, and electric kettle all into one extension cord if it has three slots.

ASSIGNMENT

1. **Drawing:** Draw a warning sign for a chemical storage cupboard. It should have a clear symbol (like a skull or an exclamation mark) and the words "DANGER - KEEP AWAY FROM CHILDREN."
2. **Rule List:** Write down three safety rules for using an iron at home.
3. **Investigation (with an adult):** Look at the label of a household cleaner (like detergent or bleach). Find and write down **one** safety instruction from the label (e.g., "Keep out of reach of children," "Use in a well-ventilated area").

WEEK 5 & 6: MIDTERM EXAMINATION AND BREAK



WEEK 7: MAINTENANCE OF HOME ECONOMICS TOOLS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Describe how to clean kitchen equipment properly.
2. Explain how to care for sewing equipment.
3. State the importance of sharpening knives.
4. Mention two ways to store tools properly.

CONTENT

1. Cleaning and Caring for Kitchen Equipment

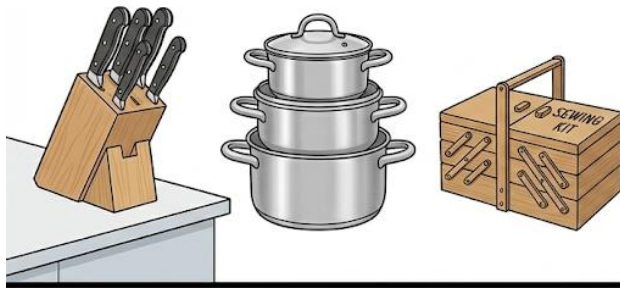
Proper cleaning keeps tools hygienic and functional.

- **Pots and Pans:** Wash with **hot, soapy water** and a sponge or scrubber after every use. For burnt-on food, soak in warm water first. **Dry completely** before stacking or putting away to prevent rust and germs.
- **Wooden Tools (Spoons, Cutting Boards):** Wash by hand, don't soak. Dry immediately to prevent warping and cracking. Occasionally rub with a little vegetable oil to keep the wood conditioned.
- **Electrical Appliances (Blender, Mixer): Unplug first!** Wash removable parts (jugs, blades, beaters) with soapy water. Wipe the motor base with a **damp cloth only**—never submerge it in water.
- **Knives:** Hand wash and dry immediately. **Do not leave soaking in water**—it dulls the blade, rusts the metal, and is a safety hazard.

2. Sharpening Knives – A Safe Edge is a Sharp Edge

- **Why Sharpen?** A **dull knife** is more dangerous than a sharp one. It requires more force to cut, which increases the chance of slipping and causing a serious injury. A sharp knife cuts cleanly and predictably.
- **How it's Done:** Sharpening is usually done by adults using a **sharpening steel** (a long rod) or a **whetstone**. The blade is drawn across the steel or stone at a specific angle. (Pupils should not attempt this without direct adult supervision).

CORRECT STORAGE



CORRECT ✓

SHARP KNIFE

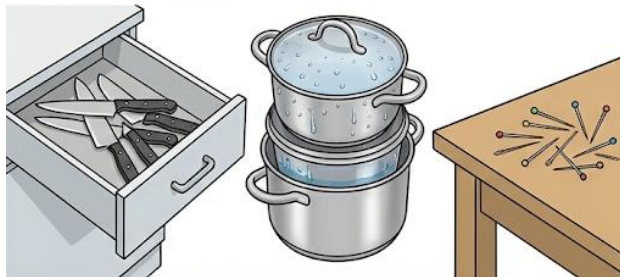


DULL KNIFE



WHY SHARP IS SAFER ✦

INCORRECT ✗



3. Caring for Sewing Equipment

- **Scissors:**
 - Use fabric scissors **only for fabric**. Cutting paper, plastic, or cardboard will dull the blades quickly.
 - Keep them dry and clean. Occasionally, an adult can put a drop of oil on the screw to keep the action smooth.
- **Needles and Pins:** Keep them in a pincushion to prevent loss and rust. A dry pincushion with a little sand or sawdust inside can help clean light rust from pins.
- **Tape Measure:** Roll it up neatly after use. Don't let it twist or knot, as this can stretch and damage it.

4. Proper Storage – A Place for Everything

Organised storage extends tool life and prevents accidents.

- **Kitchen:** Store heavy pots on lower shelves. Use a **knife block, magnetic strip, or drawer organizer** for knives. Hang utensils on hooks.
- **Sewing:** Keep your sewing kit in a box or basket where all items are contained. This prevents needles and pins from getting lost.

- **General:** Store all tools in a **clean, dry place**. Keep them away from dampness and direct sunlight.

EVALUATION

1. Why should you dry a wooden cutting board immediately after washing it?
2. Why is it dangerous to leave a kitchen knife soaking in a sink full of water?
3. What is the main safety reason for keeping knives sharp?
4. What happens to fabric scissors if you use them to cut cardboard?
5. Why is a magnetic strip a good way to store kitchen knives?
6. True or False: It's okay to store a wet blender jug back on the motor base.

ASSIGNMENT

1. **Practical Care:** Choose one kitchen tool (like a wooden spoon or a plastic bowl). Wash it, dry it thoroughly, and put it away in its correct place.
2. **Drawing:** Design an organised storage drawer for kitchen utensils. Draw the drawer and show where you would place spoons, knives (in a holder), and other tools.
3. **Interview:** Ask an adult in your home who sews or a local tailor: "How do you keep your scissors sharp and your needles from rusting?" Write down their answer.

WEEK 8: PERSONAL GROOMING AND USE OF BASIC COSMETICS I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define personal grooming.
2. Mention three steps in basic skin care.
3. Explain how to care for the hair.
4. Describe proper nail care.

CONTENT

1. What is Personal Grooming?

Personal grooming means **taking care of your own body and appearance to stay clean, healthy, neat, and presentable**. It is about **self-respect** and showing respect to others. A well-groomed person feels confident and makes a good impression.



2. Skin Care Basics – Your Body's Largest Organ

Your skin protects you from germs and regulates body temperature. It deserves care.

- **Step 1: Cleansing (Bathing)**

- **Why:** To wash away sweat, dirt, dead skin cells, and germs.
- **How:** Bathe **at least once daily** with clean water and soap. Scrub gently all over your body. Pay attention to folds (armpits, behind ears, between toes).

- **Step 2: Moisturising**

- **Why:** To keep skin soft, supple, and prevent dryness, cracking, or "ashiness."
- **How:** After bathing, while skin is slightly damp, apply a **body lotion, cream, or natural oil** (like shea butter or coconut oil). This locks in moisture.

- **Step 3: Protection**

- **Why:** To prevent damage from the sun and environment.
- **How:** Wear clean clothes. In very hot sun, wear a hat or use mild sunscreen.

3. Hair Care – Your Crowning Glory

Clean, neat hair is part of good grooming.

- **Washing:** Wash hair regularly with **shampoo** suitable for your hair type to remove dirt, oil, and prevent lice. Rinse thoroughly.
- **Combing/Brushing:** Comb or brush hair daily to keep it tidy, untangled, and to spread natural oils. Use your own comb to prevent sharing infections.
- **Neat Styling:** Hair should be kept neat and out of the face. This might mean regular trimming for boys, and tidy plaits, braids, or ponytails for girls.

4. Nail Care – Tiny Places for Germs to Hide

- **Why it's Important:** Long, dirty nails harbour germs and dirt that can easily get into your mouth when you eat, causing illness.
- **Proper Care:**
 - a. Keep fingernails and toenails **short and clean**.
 - b. Use a **nail clipper** or small scissors to trim nails straight across. **Do not bite** your nails.

- c. Clean under nails with a nail brush or the tip of a nail file while bathing.
- d. Dry feet thoroughly after washing to prevent fungal infections.

EVALUATION

1. How does good personal grooming show respect for others?
2. What is the purpose of applying lotion or oil after bathing?
3. Why is it important to use your own personal comb or hairbrush?
4. What are two health risks of having long, dirty fingernails?
5. What is the proper tool to use for trimming nails?
6. True or False: You only need to bathe when you are going to a special event.

ASSIGNMENT

1. **Routine List:** Write down your personal grooming routine for getting ready for school in the morning (e.g., 1. Brush teeth, 2. Wash face...).
2. **Drawing:** Draw two pictures side by side: one of a well-groomed pupil and one of a poorly-groomed pupil. Label two differences.
3. **Practical:** Tonight, check your fingernails. Are they short and clean? If they need trimming, ask an adult to help you or supervise you while you trim them safely.

WEEK 9: PERSONAL GROOMING AND USE OF BASIC COSMETICS II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define cosmetics.
2. List three examples of simple cosmetics.
3. Differentiate between natural and chemical beauty products.
4. Mention the importance of personal hygiene products.

CONTENT

1. What are Cosmetics?

Cosmetics are products that people apply to their **face, skin, hair, or body to clean, beautify, or change their appearance**. For primary school pupils, we focus on **simple, safe products** used for cleanliness and basic care, not heavy makeup.



2. Simple and Safe Cosmetics for Young People

- **Soap/Body Wash:** Used for **cleansing** the skin. The most important cosmetic for everyone.
- **Lotion/Cream/Moisturiser:** Used to **soften and protect** the skin from dryness.

- **Lip Balm:** A waxy product applied to lips to **prevent and treat dry, chapped, or cracked lips**, especially in harmattan.
- **Petroleum Jelly (Vaseline):** A thick jelly used to **moisturise** very dry areas like elbows, knees, and lips.
- **Deodorant:** A product (stick, roll-on, spray) applied under the arms to **reduce body odour (smell)** by preventing bacterial growth on sweat. Important as children approach adolescence.

3. Natural vs. Chemical Products

- **Natural Products:** Made from ingredients found in nature, often with little processing.
 - **Examples:** Shea butter (ori), Coconut oil, Palm kernel oil, Aloe Vera gel, Black soap (ose dudu).
 - **Advantages:** Usually gentler on the skin, less likely to cause reactions. They are part of our Nigerian heritage.
- **Chemical/Artificial Products:** Made in factories using synthetic (man-made) chemicals.
 - **Examples:** Many branded lotions, perfumes, coloured lipsticks.
 - **Caution:** Some can be harsh. **Never use skin bleaching creams** as they contain dangerous chemicals that can damage the skin permanently.

4. Self-Esteem and Healthy Choices

- **True Beauty** comes from **good health, cleanliness, confidence, and a kind character**. Cosmetics should **enhance** natural cleanliness, not hide dirt or create a false appearance.
- **Hygiene Products are Essential:** Soap, deodorant, and lotion are important for **social well-being**—they help you feel fresh and prevent offending others with body odour.
- **Be Content:** Pupils should be happy with their natural skin colour and features. Celebrate Nigerian beauty!

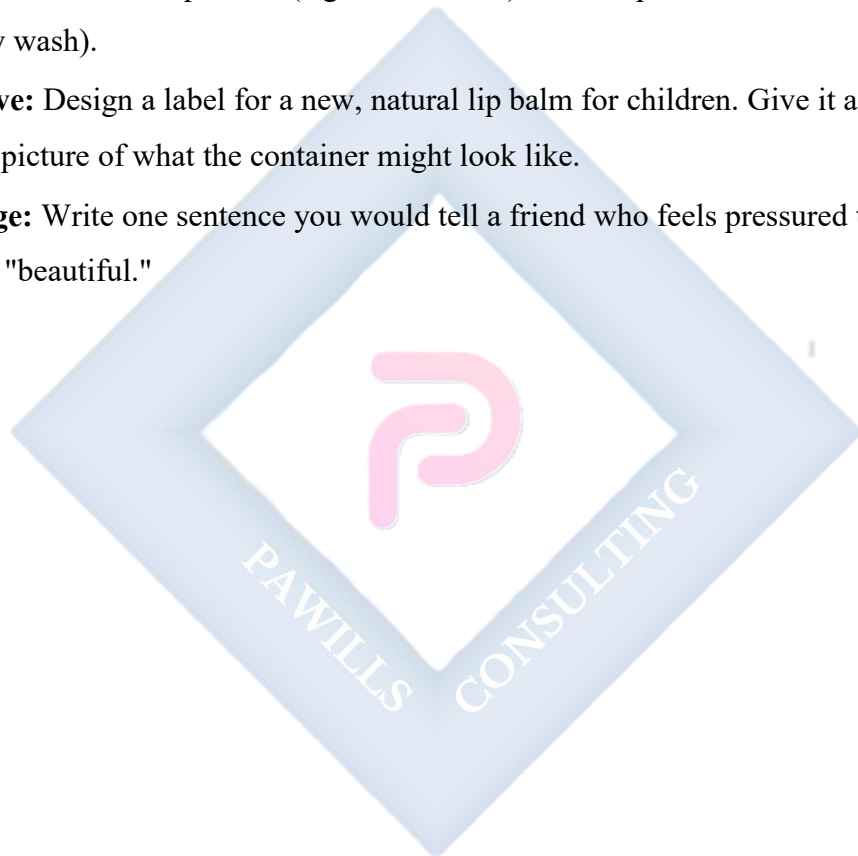
EVALUATION

1. What is the most important cosmetic product for everyone, and why?
2. When would it be most useful to apply lip balm?

3. Name two natural Nigerian products that can be used to moisturise the skin.
4. Why are skin bleaching creams extremely dangerous?
5. What is the main purpose of using a deodorant?
6. True or False: Wearing expensive makeup is more important than being clean.

ASSIGNMENT

1. **Product Hunt:** Look at the products in your bathroom at home. Find and write down the names of **one** natural product (e.g., Shea butter) and **one** product used for cleansing (soap or body wash).
2. **Creative:** Design a label for a new, natural lip balm for children. Give it a name, and draw a picture of what the container might look like.
3. **Message:** Write one sentence you would tell a friend who feels pressured to use makeup to look "beautiful."



WEEK 10: INTRODUCTION TO ENTREPRENEURSHIP

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define entrepreneurship.
2. Mention who an entrepreneur is.
3. List three skills that can be turned into a business.
4. Explain how creativity helps a business.

CONTENT

1. What is Entrepreneurship?

Entrepreneurship is the **process of starting and running your own business to solve a problem or meet a need, with the goal of making a profit (money).** It involves **taking a risk** (using time and money) in the hope of a reward.



2. Who is an Entrepreneur?

An **entrepreneur** is a person who **sees an opportunity, creates a plan, and works hard to start and manage a business**. They are **leaders, innovators, and problem-solvers**.

- **Examples in your community:** The woman who sells *akara* by the roadside, the man with a barbing salon, the person who repairs phones, the farmer who sells his vegetables.

3. Turning Skills into Business – You Can Be One!

The skills you learn in Pre-Vocational Studies are perfect for starting a small business.

- **From Agriculture:**
 - Growing and selling vegetables (lettuce, tomatoes) from a small garden.
 - Rearing and selling chickens or eggs.
 - Making and selling organic manure/compost.
- **From Home Economics:**
 - Baking and selling snacks (chin-chin, cake, puff-puff).
 - Making and selling liquid soap or air freshener.
 - Mending clothes (sewing buttons, fixing hems) for a small fee.
 - Making bead necklaces or bracelets.

4. Creativity and Innovation – Making Your Business Special

- **Creativity** is thinking of **new and original ideas**.
- **Innovation** is **improving** something that already exists.
- **How they help a business:** They make your product or service **stand out** from others. If everyone sells plain *chin-chin*, you could sell **spicy *chin-chin* or chocolate *chin-chin***. If everyone sells water in sachets, you could sell **flavoured water with a slice of lemon** in a nice bottle. Creativity attracts customers!

5. Key Entrepreneurial Qualities

- **Hard work:** A business needs effort.
- **Honesty:** Build trust with customers.
- **Persistence:** Don't give up if things are slow at first.

- **Money Management:** Keep track of what you spend (cost) and what you earn (income) to know your profit.

EVALUATION

1. What is the main goal of an entrepreneur when starting a business?
2. Look around your community. Name one entrepreneur and describe what they sell or do.
3. Which subject, Agriculture or Home Economics, teaches you skills that could be used to start a *chin-chin* selling business?
4. If you are good at drawing, what is one simple business you could start at school?
5. How can being creative help you sell more bottles of water than the next person?
6. True or False: Only adults can be entrepreneurs.

ASSIGNMENT

1. **Business Brainstorm:** Think of a simple business you could start with less than ₦500. Write down: a) Your business name. b) What you will sell. c) Who will buy it (your customers).
2. **Interview:** Talk to a small business owner in your area (a shopkeeper, a tailor). Ask them: "What is the most important thing you had to learn when you started?" Write their answer.
3. **Advertisement Design:** Create a colourful sign or jingle (short song) to advertise your imaginary business from question 1. What makes it special?

WEEK 11: REVISION

1. Comprehensive Review of Term 3 Topics

Let's revisit the key areas we covered this term:

- **Home Economics Tools (Weeks 1-2 & 7):** Identifying and caring for kitchen, sewing, and cleaning tools. Safe storage.
- **Safety (Weeks 3-4):** Rules for handling knives, irons, chemicals, and preventing fires. First aid awareness.
- **Personal Grooming (Weeks 8-9):** Daily routines for skin, hair, and nail care. Understanding simple, safe cosmetics and the dangers of skin bleaching.
- **Entrepreneurship (Week 10):** What it means to be an entrepreneur; turning skills into small businesses; the importance of creativity and hard work.

2. Interactive Revision Activities

- **"Toolbox Challenge":** The teacher shows a picture of a tool (e.g., a knife, an iron, a measuring cup). Pupils must state its name, one use, and **one safety rule** for using it.
- **"Grooming Routine Race":** In teams, pupils put pictures in the correct order for a morning grooming routine (e.g., Wake up → Bathe → Apply lotion → Brush hair → Put on clean clothes).
- **Entrepreneurial Pitch:** In small groups, pupils have 2 minutes to "pitch" a simple business idea based on a skill learned this term. The class votes on the most creative idea.
- **Safety Scenario Quiz:** "What would you do if...?" scenarios read by the teacher, with multiple-choice answers.

EVALUATION *(A final consolidation before examinations.)*

1. You want to measure half a cup of oil for frying. Which tool do you use, and what is one safety tip for the kitchen when you are about to start frying?
2. Describe two differences between a natural cosmetic (like shea butter) and a chemical skin bleaching cream.

3. Your friend wants to start a business selling handmade cards. Which Pre-Vocational skills would they be using? What is one way they could be creative with their cards to attract more buyers?
4. Why is it dangerous to leave an iron lying flat on the ironing board while it is still plugged in?
5. What does the word "entrepreneur" mean, and can you name one quality a good entrepreneur needs?
6. Put these grooming steps in order: (Wear clean clothes, Bathe with soap, Apply lotion).

ASSIGNMENT

1. **Study Aid:** Create a "Term 3 Top 5" list. Write down the five most important things you learned this term, with a small drawing next to each.
2. **Teach a Skill:** Choose one simple safety or grooming rule from this term. Teach it to a younger sibling or a friend. Write down what you taught and who you taught it to.
3. **Final Preparation:** Ensure all your notes for Weeks 1-10 are complete. Write one question about any topic you would like clarified before the exams.